

Biostatistics

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June, 2023

Objectives

General Objective

- ◆ The aim of biostatistical study is to provide the numbers that contain information about certain situations and present them in such a way that valid interpretations are possible

Specific Objectives

- ◆ interpret summary statistics, graphical displays and contingency tables commonly presented in the health literature
- ◆ carry out exploratory data analysis

- **Descriptive statistics**

Methods of data organization and presentation.

Methods of data presentation

Measures of relative standing

Measures of position

Presenting and summarizing data

Describing categorical variables

- ◆ **Table of frequency distributions**
 - Frequency
 - Relative frequency
 - Cumulative frequencies
 - Relative cumulative frequency
- ◆ **Diagrams and Charts**
 - Bar charts
 - Pie charts
 - Pictogram
 - Histogram
 - Frequency polygon
 - Ogive

Table of frequency distributions

Guidelines for constructing tables

- ◆ Keep them simple
- ◆ All tables should be self-explanatory
- ◆ Include **clear title** telling **what, when and where**
- ◆ Clearly label the rows and columns
- ◆ State clearly the unit of measurement used
- ◆ Explain codes and abbreviations in the foot-note
- ◆ Show totals
- ◆ If data is not original, indicate the source in foot-note.

Frequency Distribution: The organization of raw data in table form with classes and frequencies.

Categorical Frequency distributions

- ◆ Simple and effective way of summarizing **categorical data**
- ◆ Done by counting the number of observations falling into each of the categories or levels of the variables.

E.g. birth weight with levels ‘**Very low**’, ‘**Low**’, ‘**Normal**’ and ‘**big**’.

- ◆ The frequency distribution for newborns is obtained simply by counting the number of newborns in each birth weight category.

Relative Frequency

- ◆ It is the proportion or percentages of observations in each category.
- ◆ The distribution of proportions is called the *relative frequency* distribution of the variable
- ◆ Given a total number of observations, the relative frequency distribution is easily derived from the frequency distribution.
- ◆ Conversion in the opposite direction is also possible, but the conversion is often inaccurate because of rounding

Cumulative frequency

- ◆ It is the number of observations in the category plus observations in all categories smaller or greater than it.

Cumulative relative frequency

- ◆ It is the proportion of observations in the category plus observations in all categories smaller than or greater than it.
- ◆ It is obtained by dividing the cumulative frequency by the total number of observations.

Table 1. Distribution of birth weight of newborns between 1976-1996 at AA.

BWT	Freq.	Cum. Freq	Rel.Freq(%)	Cum.rel.freq.(%)
Very low	43	43	0.4	0.4
Low	793	836	8.0	8.4
Normal	8870	9706	88.9	97.3
Big	268	9974	2.7	100
Total	9974		100	

Con...

◆ **Ungrouped frequency Distribution:**

It is a table of all the potential raw score values that could possibly occur in the data along with the number of times each actually occurred. It is often constructed for small set or data on discrete variable.

◆ **Grouped frequency distribution**

When the range of the data is large, the data must be grouped in to classes that are more than one unit in width.

◆ **Grouped Frequency Distribution:** A frequency distribution where several numbers are grouped into one class.

◆ Select a set of continuous, non-overlapping intervals such that each value can be placed in one and only one of the intervals.

Example:

Leisure time (hours) per week for 40 college students:

23 24 18 14 20 36 24 26 23 21 16 15 19 20 22 14 13
10 19 27 29 22 37 28 34 32 23 19 21 31 16 28 19 18
12 27 15 21 25 16

- ◆ $L=37, S=10$
- ◆ $R=37-10=27$
- ◆ $K = 1 + 3.322 (\log 40) = 6.32 \approx 7$
- ◆ $\text{Width} = (37-10)/7 = 3.9 \approx 4$

Example:

- ◆ Let us take the starting point as 10
- ◆ the lower class limits will be 10,14,18,22,26,30,34
- ◆ The upper class limits are 13,17,21,25,29,33,37

Time (Hours)	Class boundary	Mid point	Frequency	Relative Frequency	Cumulative Relative Frequency
10-13	9.5-13.5	11.5			
14-17	13.5-17.5	15.5			
18-21	17.5-21.5	19.5			
22-25	21.5-25.5	23.5			
26-29	25.5-29.5	27.5			
30-33	29.5-33.5	31.5			
34-37	33.5-37.5	35.5			
Total			40	1.00	

Exercise

- ◆ Construct a grouped frequency distribution for the following data.

11	29	6	33	14	31	22	27	19	20
18	17	22	38	23	21	26	34	39	27

Solution

- ◆ Step 1: Find the highest and the lowest value $H=39$, $L=6$
- ◆ Step 2: Find the range; $R=H-L=39-6=33$
- ◆ Step 3: Select the number of classes' desired using Sturges formula;
$$k = 1 + 3.32 \log n = 1 + 3.32 \log (20) = 5.32 = 6 \text{ (rounding up)}$$
- ◆ Step 4: Find the class width; $w=R/k=33/6=5.5=6$ (rounding up)
- ◆ Step 5: Select the starting point, let it be the minimum observation. 6, 12, 18, 24, 30, 36 are the lower class limits.
- ◆ Step 6: Find the upper class limit;
e.g. the first upper class $=12-U=12-1=11$
11, 17, 23, 29, 35, 41 are the upper class limits.

Solution

- ◆ So combining step 5 and step 6, one can construct the following classes.

Class limits

6 – 11

12 – 17

18 – 23

24 – 29

30 – 35

36 – 41

- ◆ Step 7: Find the class boundaries;

E.g. for class 1 Lower class boundary = $6 - U/2 = 5.5$

Upper class boundary = $11 + U/2 = 11.5$

Then continue adding w on both boundaries to obtain the rest boundaries. By doing so, one can obtain the following classes.

Solution

Class boundary

5.5 – 11.5

11.5 – 17.5

17.5 – 23.5

23.5 – 29.5

29.5 – 35.5

35.5 – 41.5

- ◆ Step 8: tally the data.
- ◆ Step 9: Write the numeric values for the tallies in the frequency column.
- ◆ Step 10: Find cumulative frequency.
- ◆ Step 11: Find relative frequency or/and relative cumulative frequency.

The complete frequency distribution follows:

Solution

Class limit	Class boundary	Class mark	Tally	Freq.	Cf (< type)	Cf (> type)	R.f.	C.R.f. (< type)
6-11	5.5-11.5	8.5	//	2	2	20	0.10	0.10
12-17	11.5-17.5	14.5	//	2	4	18	0.10	0.20
18-23	17.5-23.5	20.5	/// //	7	11	16	0.35	0.55
24-29	23.5-29.5	26.5	///	4	15	9	0.20	0.75
30-35	29.5-35.5	32.5	///	3	18	5	0.15	0.90
36-41	35.5-41.5	38.5	//	2	20	2	0.10	1.00

Diagrammatic Representation

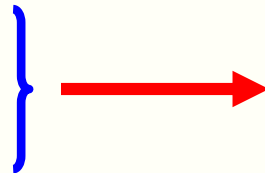
Pictorial representations of Statistical data

Importance of diagrammatic and graphic representation

1. Diagrams have greater attraction than mere figures.
2. They give quick overall impression of the data.
3. They have great memorizing value than mere figures.
4. They facilitate comparison
5. Used to understand patterns and trends

Specific types of diagrams include:

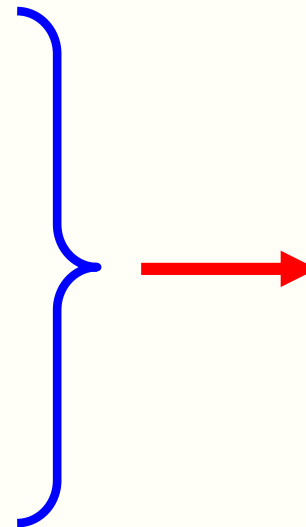
- Bar graph
- Pie chart



**Nominal, ordinal,
Discrete data**

types of graphs include:

- Histogram
- Frequency polygon
- Cum. Freq. polygon
- Line graph
- Others



**Quantitative
continuous
data**

1. Bar charts

- ◆ Categories are listed on the horizontal axis (X-axis)
- ◆ Frequencies or relative frequencies are represented on the Y-axis (ordinate)
- ◆ The height of each bar is proportional to the frequency or relative frequency of observations in that category
- ◆ All the bars must have equal width
- ◆ The bars are not joined together (leave space between bars)
- ◆ The different bars should be separated by equal distances
- ◆ All the bars should rest on the same line called the base
- ◆ Label both axes clearly
- ◆ There are different types of bar graphs.

A. Simple bar chart:

It is a one-dimensional in which the bar represents the whole of the magnitude.

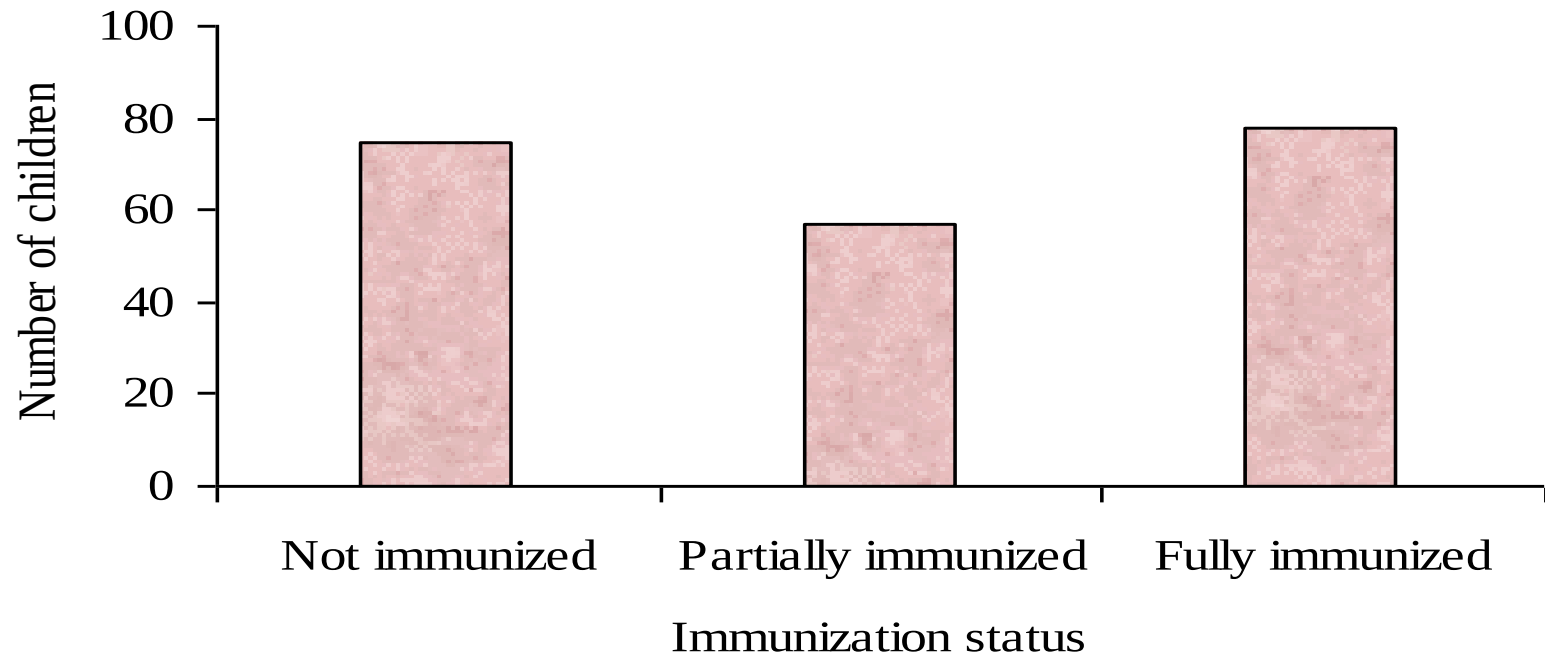
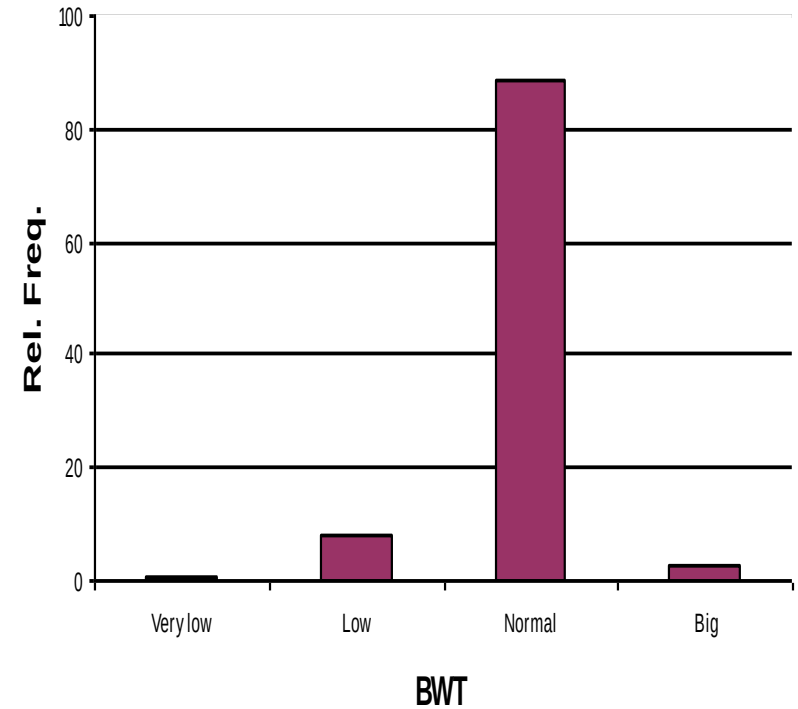
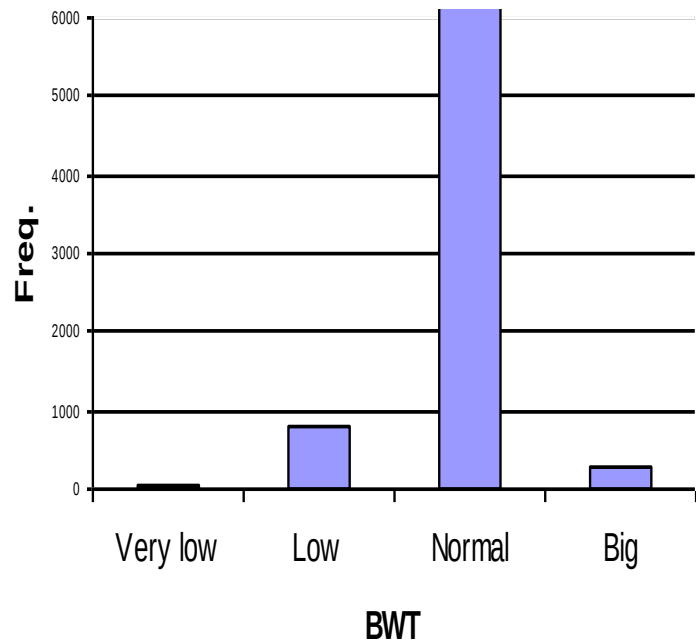


Fig. 1. Immunization status of Children in Adami Tulu Woreda, Feb. 1995

Bar charts showing frequency distribution of the variable 'BWT'



B. Multiple bar chart :

the component figures are shown as separate bars adjoining each other. It depicts distributional pattern of more than one variable

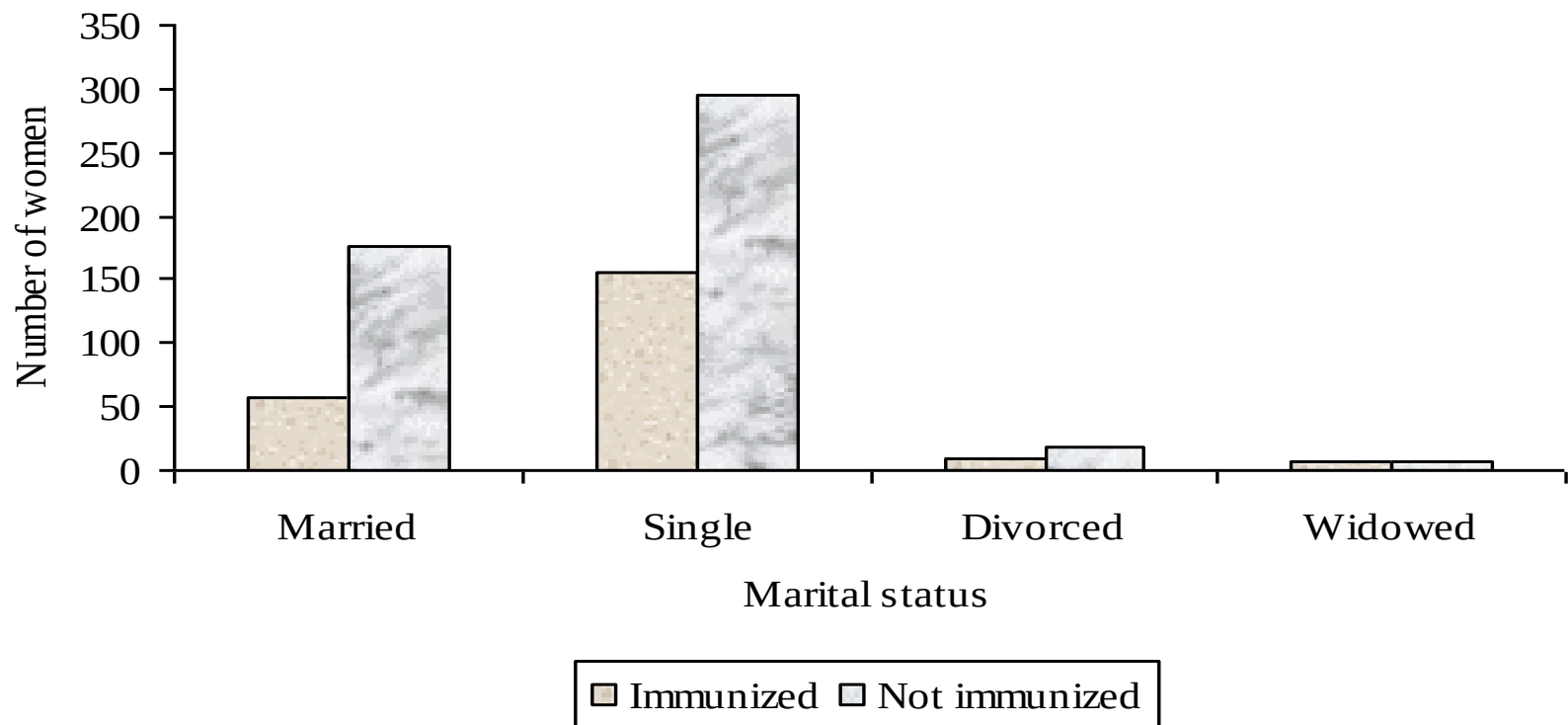
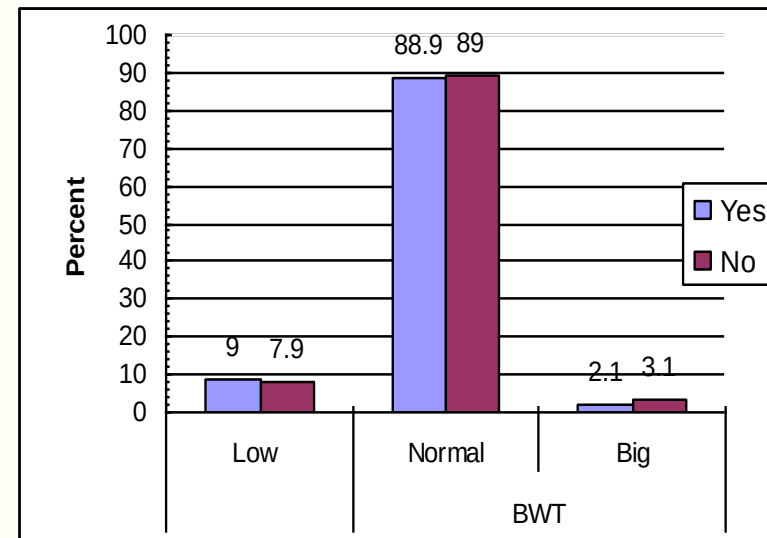
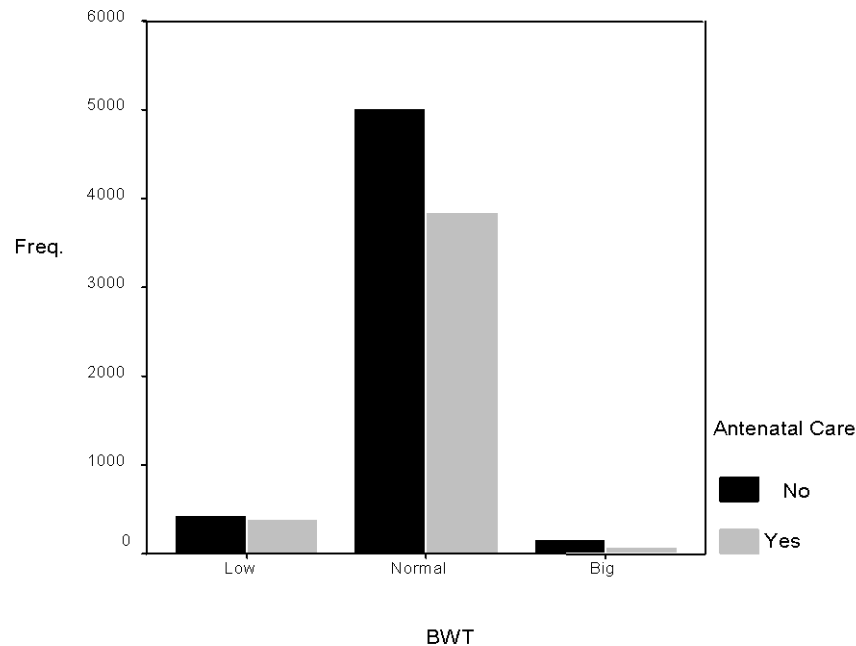


Fig. 2 TT Immunization status by marital status of women 15-49 years, Asendabo town, 1996

Bar charts for comparison

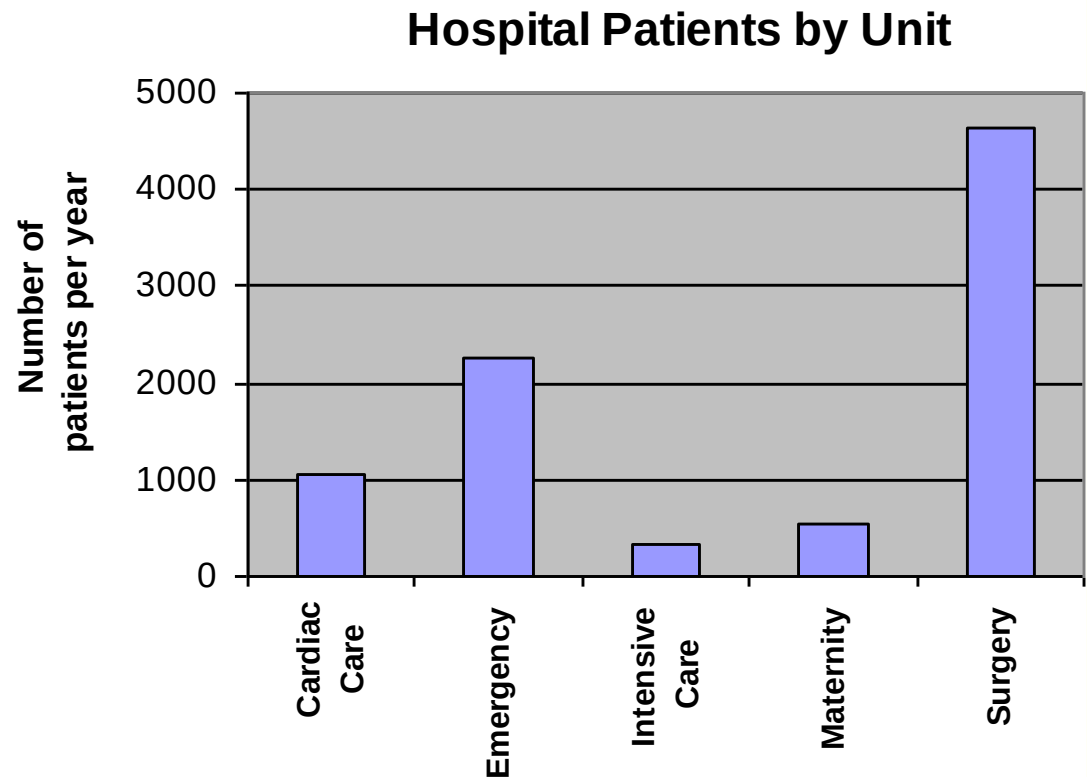
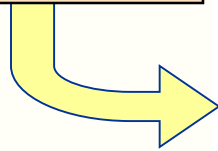
- ◆ In order to compare the distribution of a variable for two or more groups, bars are often drawn along side each other for groups being compared in a single bar chart



Bar chart indicating categories of birth weight of 9975 newborns grouped by antenatal follow-up of the mothers

Bar Chart Example

Hospital Unit	Number of Patients
Cardiac Care	1,052
Emergency	2,245
Intensive Care	340
Maternity	552
Surgery	4,630



C. Component (sub-divided) bar chart:

Bars are sub-divided into component parts of the figure. These sorts of graphs are constructed when each total is built up from two or more component figures.

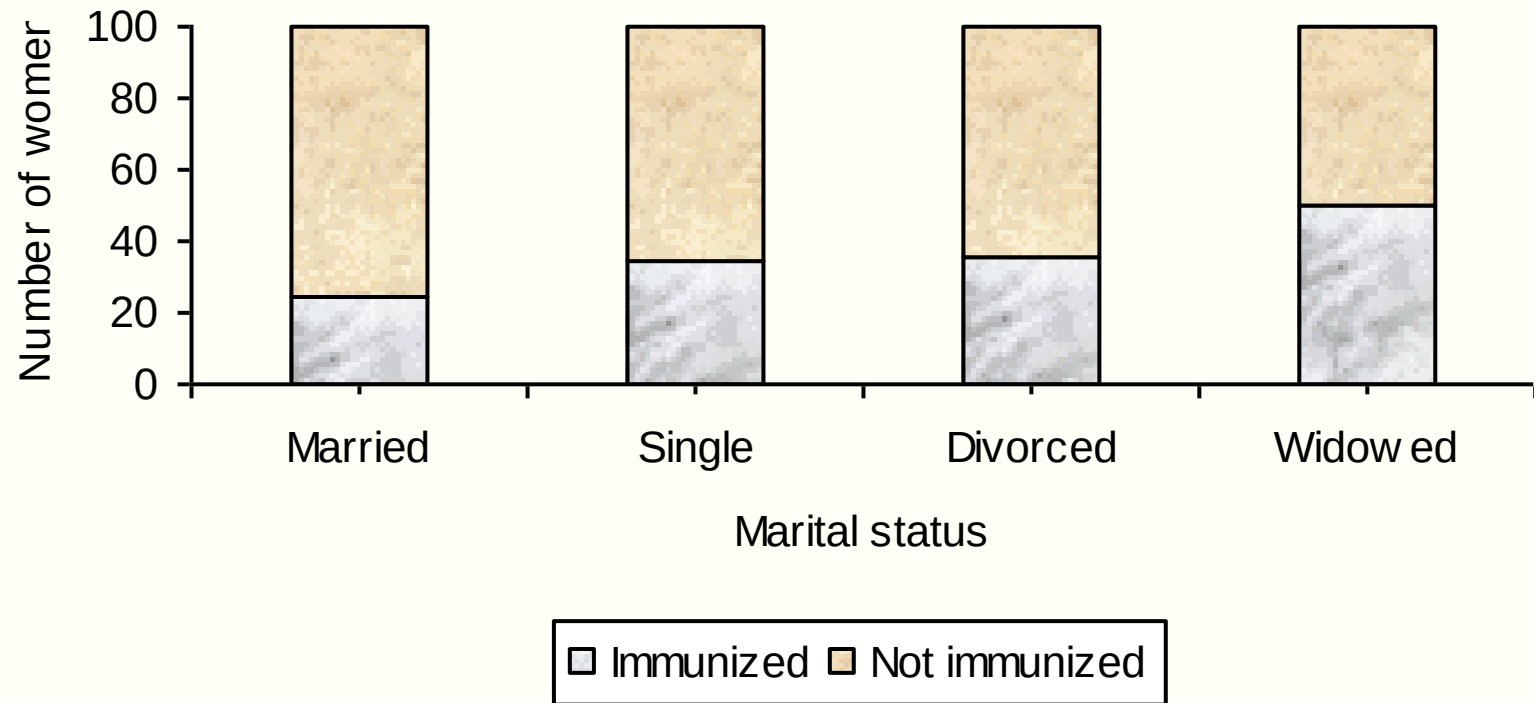
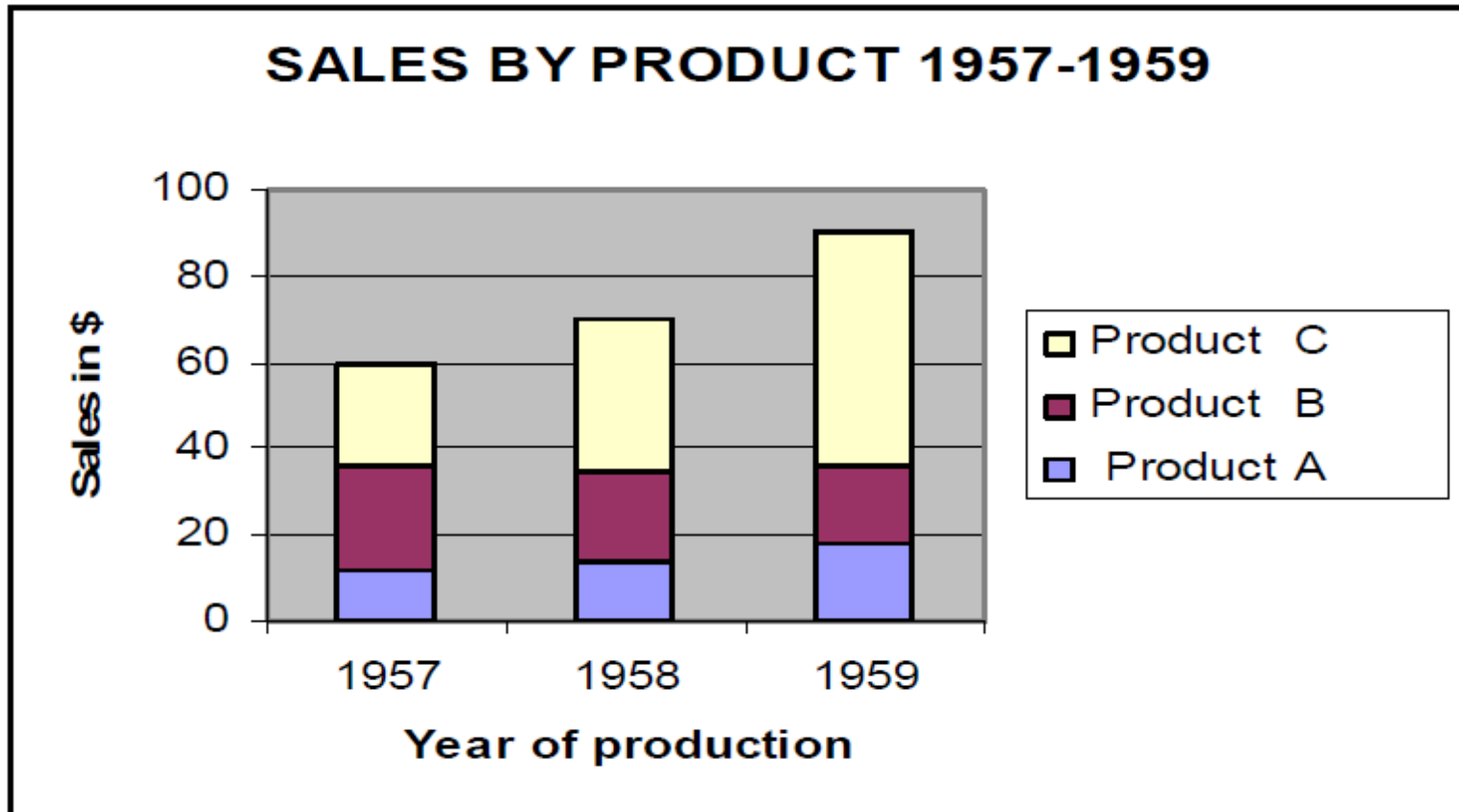


Fig. 3 TT Immunization status by marital status of women 15-49 years, Asendabo town, 1996

Component bar chart



2. Pie chart

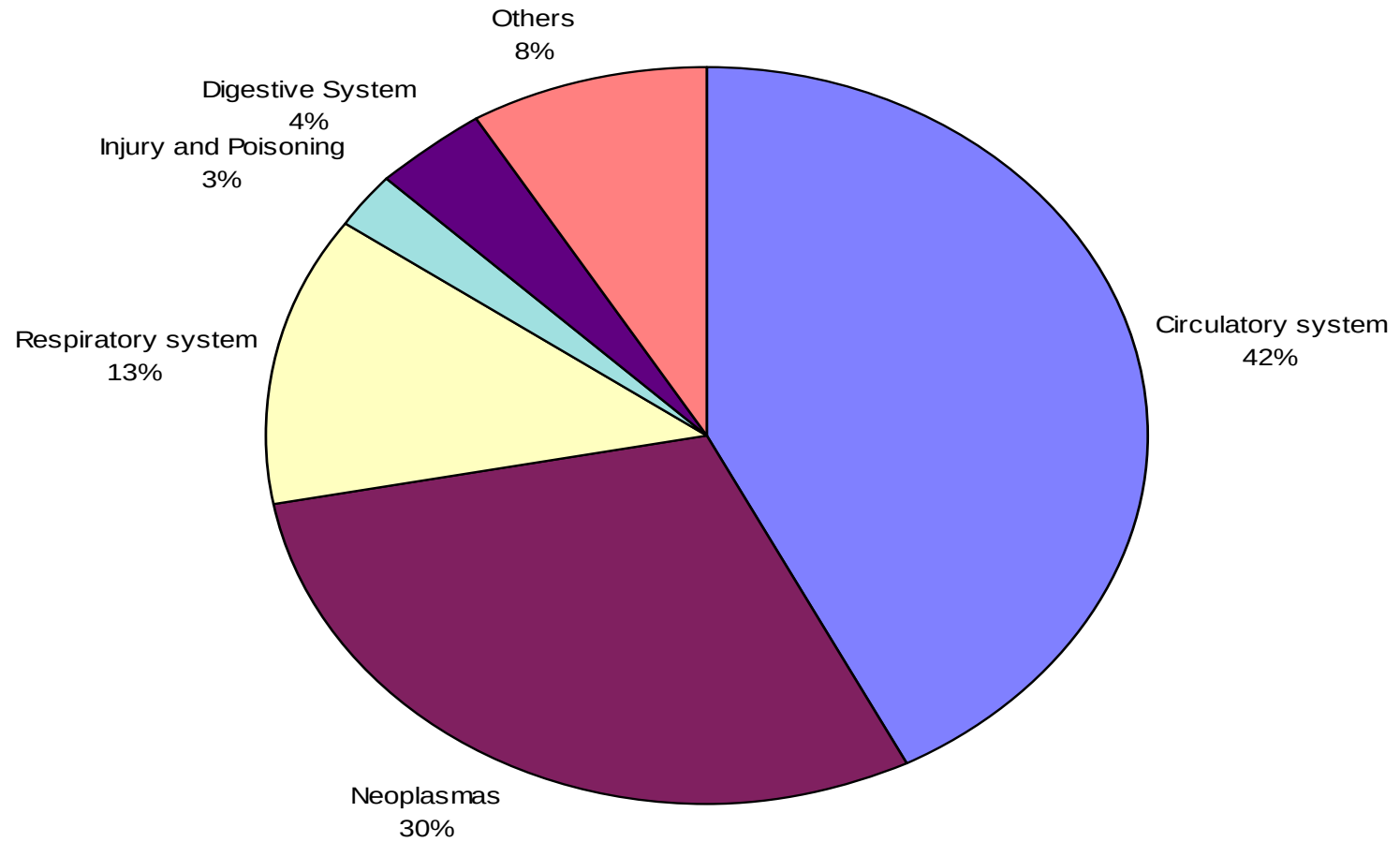
- ◆ Shows the relative frequency for each category by dividing a circle into sectors
- ◆ The angles are proportional to the relative frequency.
- ◆ Used for a single categorical variable
- ◆ Use percentage distributions

Relative Frequency	Size of Wedge, in degrees
0.50	50% of 360 = 180 degrees
0.25	25% of 360 = 90 degrees
p	$(p) \times (100\%) \times 360$ degrees

Example: Distribution of deaths for females, in England and Wales, 1989.

Cause of death	No. of death
Circulatory system	100 000
Neoplasm	70 000
Respiratory system	30 000
Injury and poisoning	6 000
Digestive system	10 000
Others	20 000
Total	236 000

Distribution fo cause of death for females, in England and Wales, 1989



Pictogram

we represent data by means of some picture symbols. We decide about a suitable picture to represent a definite number of units in which the variable is measured.

Example: Draw a pictorial diagram to present the following data (number of patients in a certain country for four years.)

Let a single picture () represents one thousand patients.

Year	1992	1993	1994	1995
No. of students	2000	3000	5000	7000

1995	      	
1994	    	Key:  = 1000 patients
1993	  	
1992	 	

Graphical representation of data

- Histogram
- Frequency polygon
- Ogive (cumulative frequency polygon)

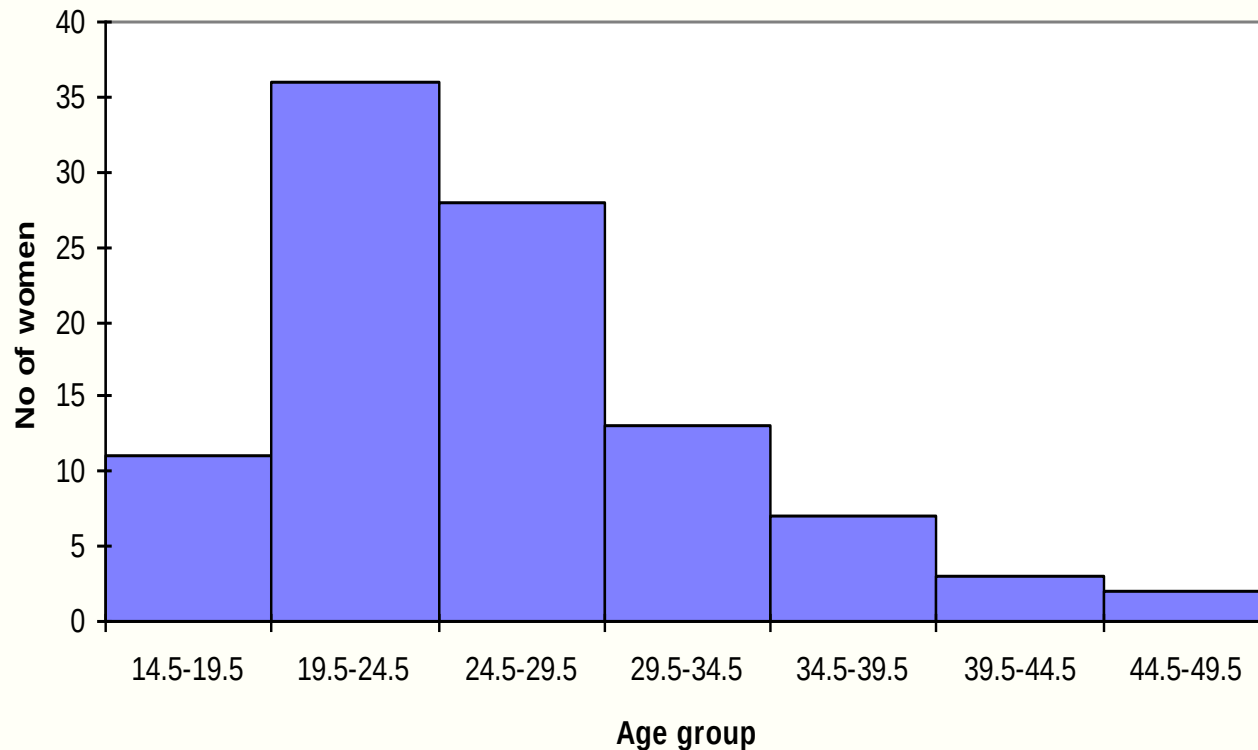
Histograms

- ◆ Histograms are frequency distributions with continuous class interval that have been turned into graphs.
- ◆ Given a set of numerical data, we can obtain impression of the shape of its distribution by constructing a histogram.
- ◆ Horizontal axis: Labels of the variable
- ◆ Vertical bar: Frequency or the relative frequency
- ◆ Except for the two boundaries, class intervals are usually chosen to be of equal width.
- ◆ If this is not the case, the histogram could give a misleading impression of the shape of the data

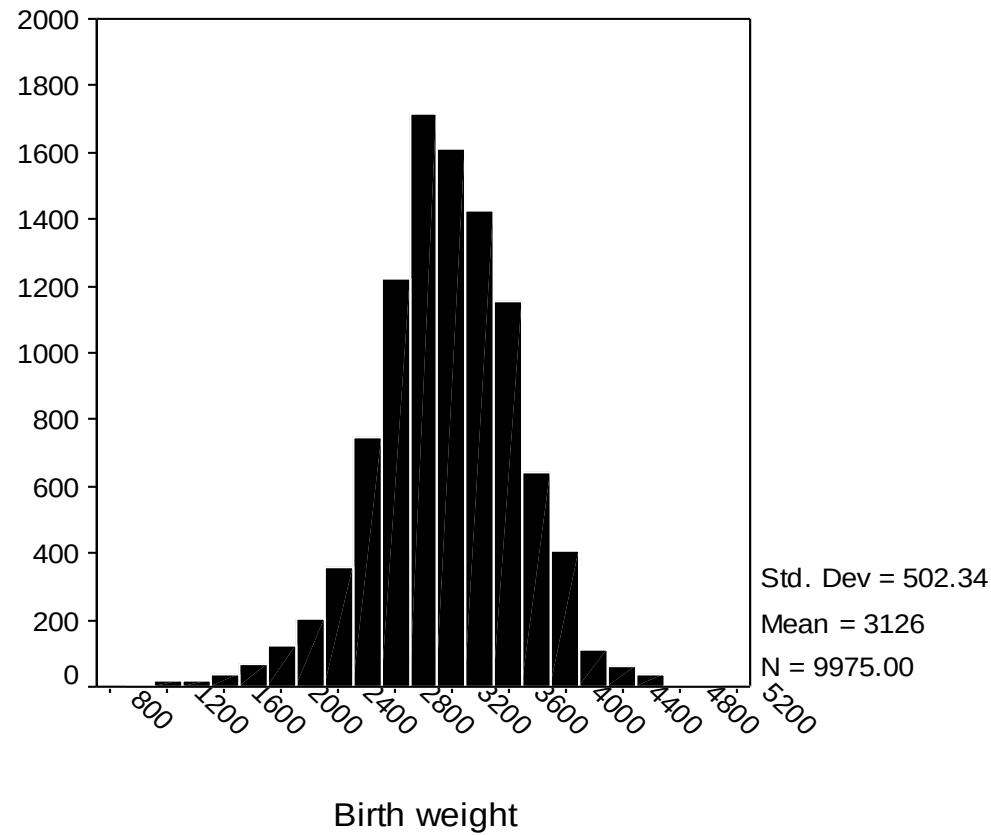
Example: Distribution of the age of women at the time of marriage

Age group	15-19	20-24	25-29	30-34	35-39	40-44	45-49
Number	11	36	28	13	7	3	2

Age of women at the time of marriage



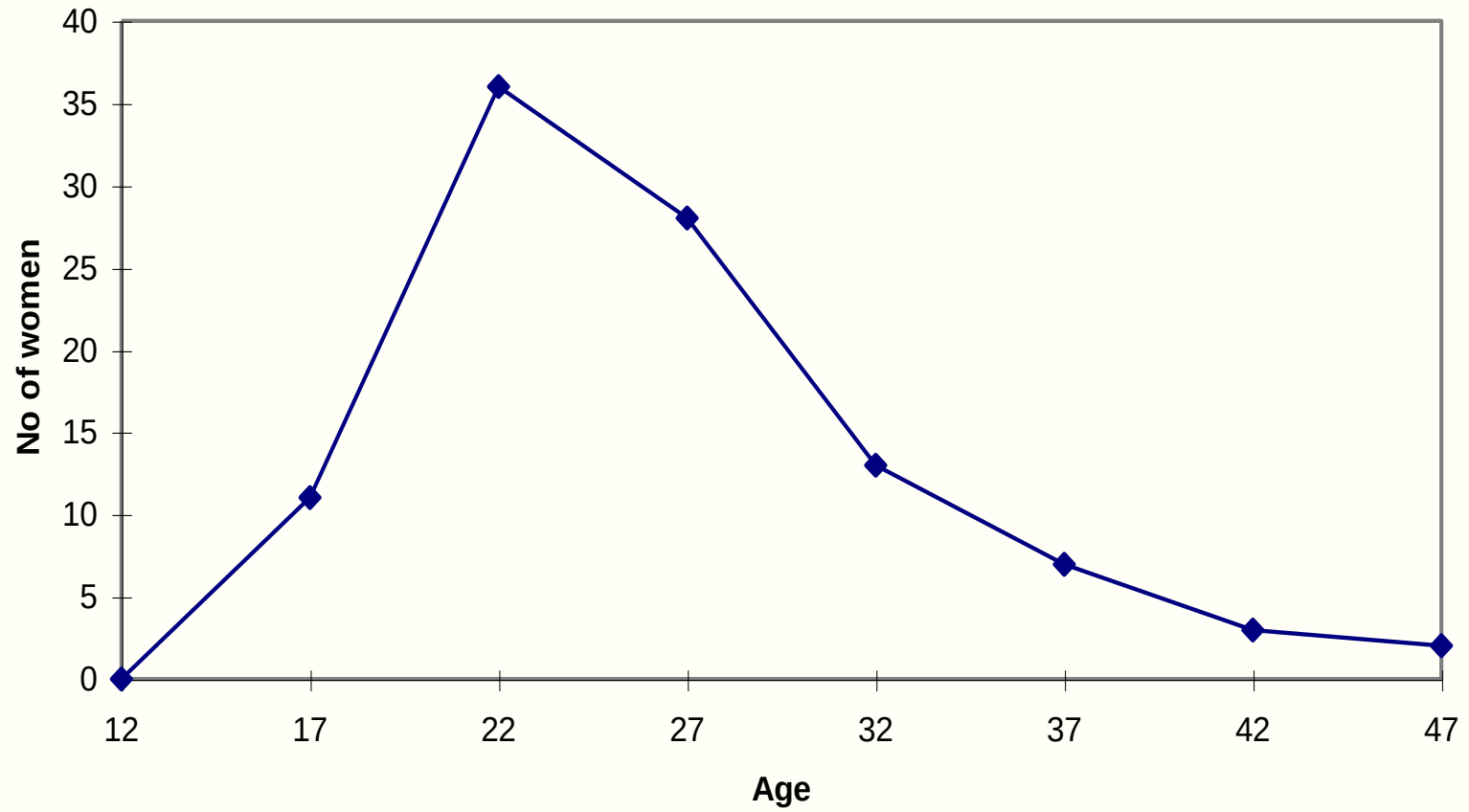
A histogram displaying frequency distribution of birth weight of newborns at Tikur Anbessa Hospital



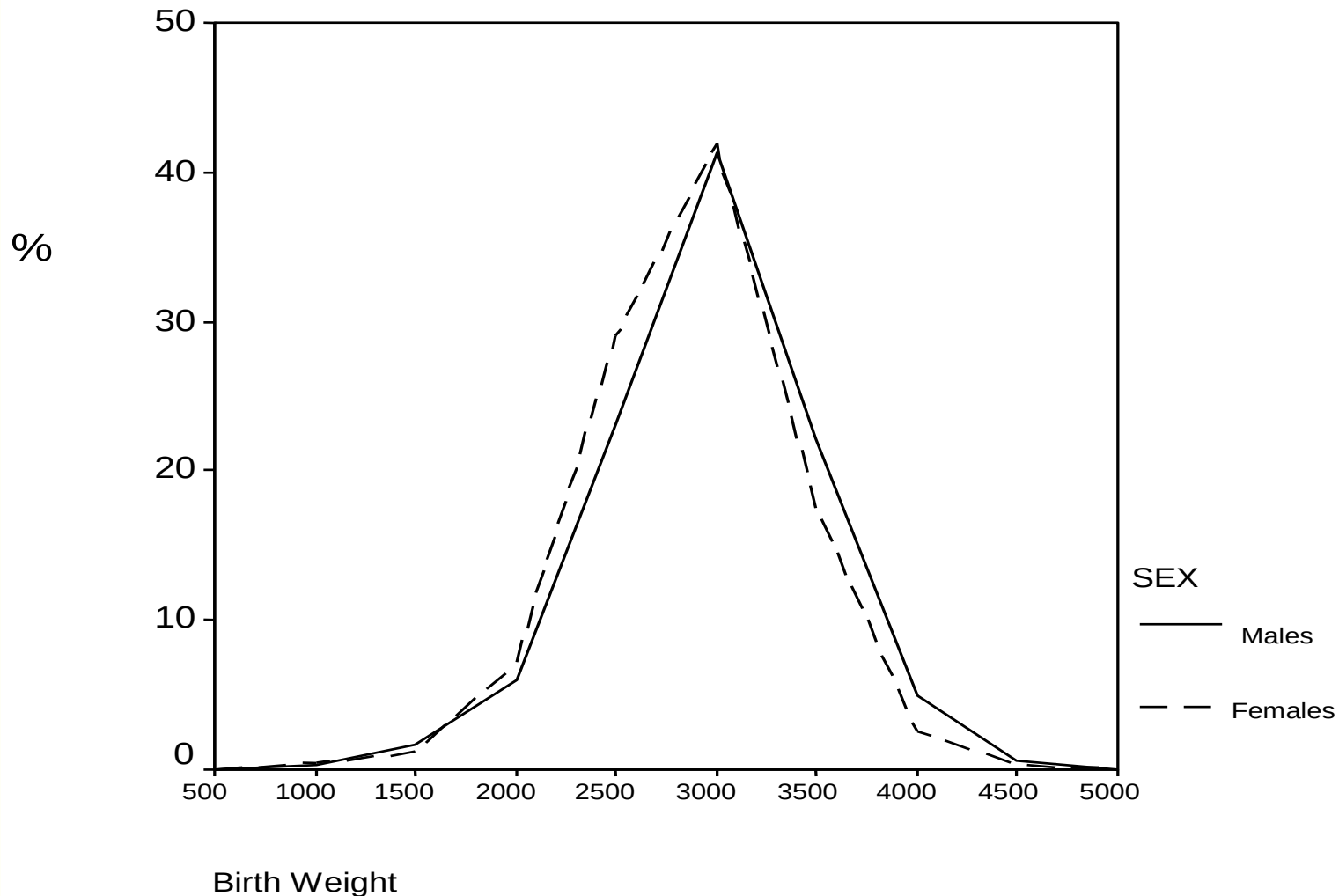
Frequency polygons

- ◆ Instead of drawing bars for each class interval, sometimes a single point is drawn at the mid point of each class interval and consecutive points joined by straight line.
- ◆ A graph drawn in this way is called frequency polygons (line graphs).
- ◆ Frequency polygons are superior to histograms for comparing two or more sets of data.

Age of women at the time of marriage



Frequency polygon of birth weight of 9975 newborns for males and females



Cumulative frequency polygons (ogive)

- ◆ Some times it may be necessary to know the number of items whose vale are more or less than a certain amount.
- ◆ For example we may be interested in knowing the number of patients whose weight is less than 50kg or more than say 60kg.
- ◆ To get this information it is necessary to change the form of frequency distribution from simple to cumulative distribution.
- ◆ Horizontal axis: Labels of the variable
- ◆ Vertical bar: cumulative relative frequency.
- ◆ The points are then connected by straight lines.
- ◆ Like frequency polygons, cumulative frequency polygons may be used to comparing sets of data.
- ◆ Cumulative frequency polygons can also be used to obtain percentiles of a set of data.

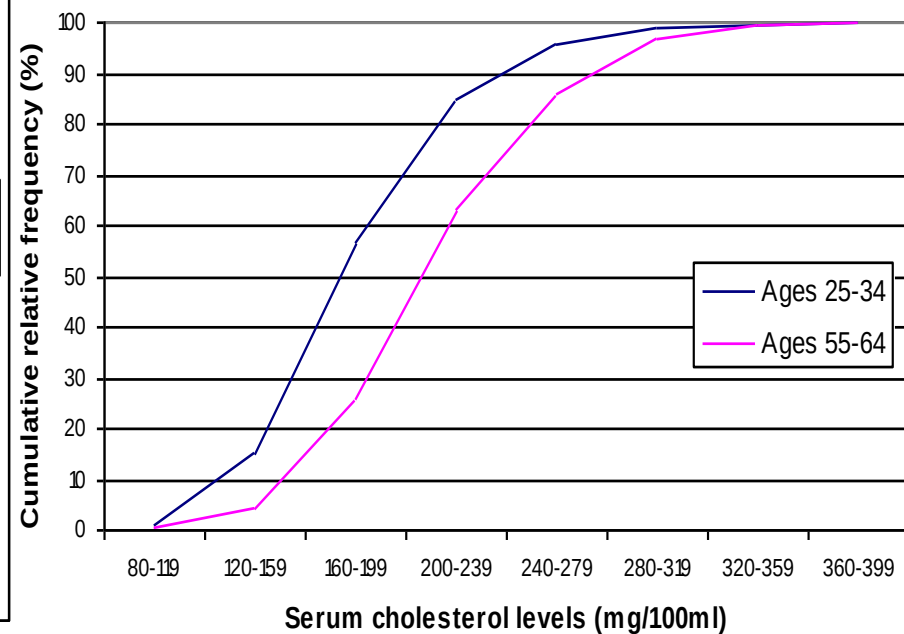
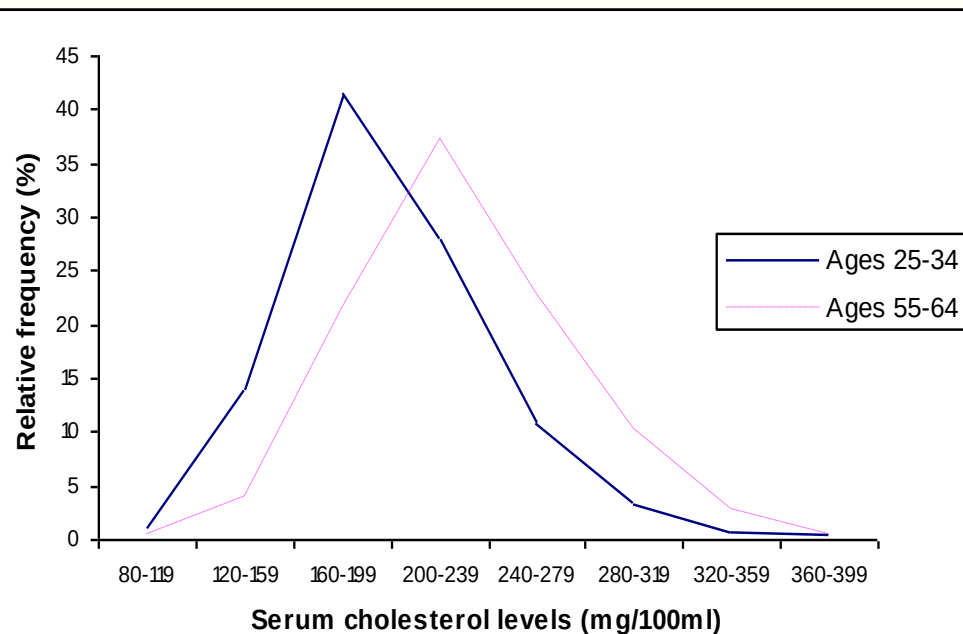
Table 1. Frequencies of serum cholesterol levels for 1067 US males of ages 25-34 1976-1980

Cholesterol level Mg/100ml	freq	Relative freq	Cum freq	Cum.rel. freq
80-119	13	1.2	13	1.2
120-159	150	14.1	163	15.3
160-199	442	41.4	605	56.7
200-239	299	28.0	904	84.7
240-279	115	10.8	1019	95.5
280-319	34	3.2	1053	98.7
320-359	9	0.8	1062	99.5
360-399	5	0.5	1067	100
Total	1067	100		

Table 2. Frequencies of serum cholesterol levels for 1227 US males of ages 55-64 1976-1980

Cholesterol level Mg/100ml	freq	Relative freq	Cum freq	Cum.rel. freq
80-119	5	0.4	5	0.4
120-159	48	3.9	53	4.3
160-199	265	21.6	318	25.9
200-239	458	37.3	776	63.2
240-279	281	22.9	1057	86.1
280-319	128	10.4	1185	96.5
320-359	35	2.9	1220	99.4
360-399	7	0.5	1227	100
Total	1227	100		

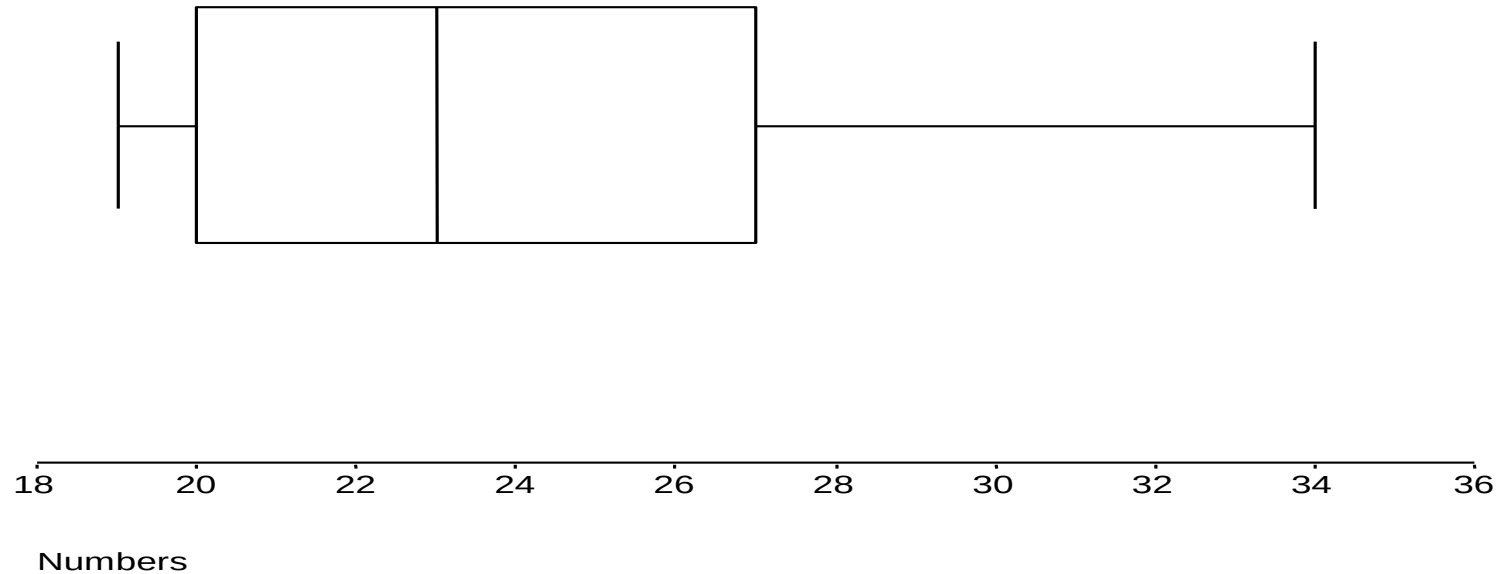
Frequency polygon and Cumulative frequency polygons of serum cholesterol levels for 2294 males aged 25-34 and 55-64 years, 1976-1980



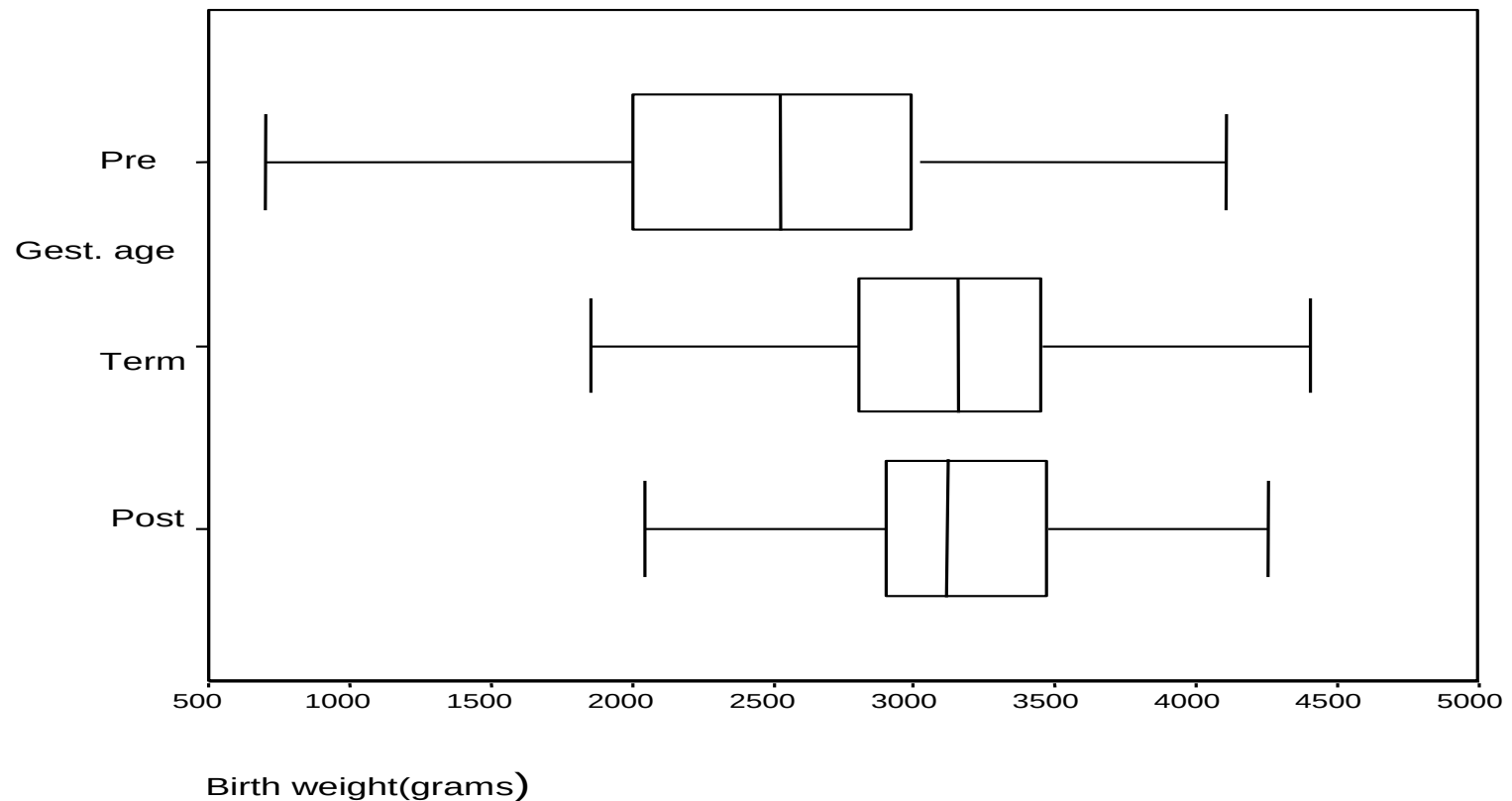
Box Plots

- ◆ A visual picture called ***box plot*** can be used to convey a fair amount of information about certain location in the distribution of a set of data.
- ◆ The box shows the distance between the first and the third quartiles,
- ◆ The median is marked as a line within the box and
- ◆ The end lines show the minimum and maximum values respectively

Illustration of Box-plot



A box-plot indicating birth weight of 5092 newborns by gestational age at Tikur Anbessa Hospital studied



Line graph

- ◆ Useful for assessing the trend of particular situation **overtime**.
- ◆ Helps for monitoring the **trend of epidemics**.
- ◆ The time, in weeks, months or years, is marked along the **horizontal axis**, and
- ◆ Values of the quantity being studied is marked on the **vertical axis**.
- ◆ Values for each category are connected by **continuous line**.
- ◆ Sometimes two or more graphs are drawn on the same graph taking the same scale so that the plotted **graphs are comparable**.

Example: Malaria Parasite Prevalence Rates in Ethiopia, 1967 – 1979 Eth. C.

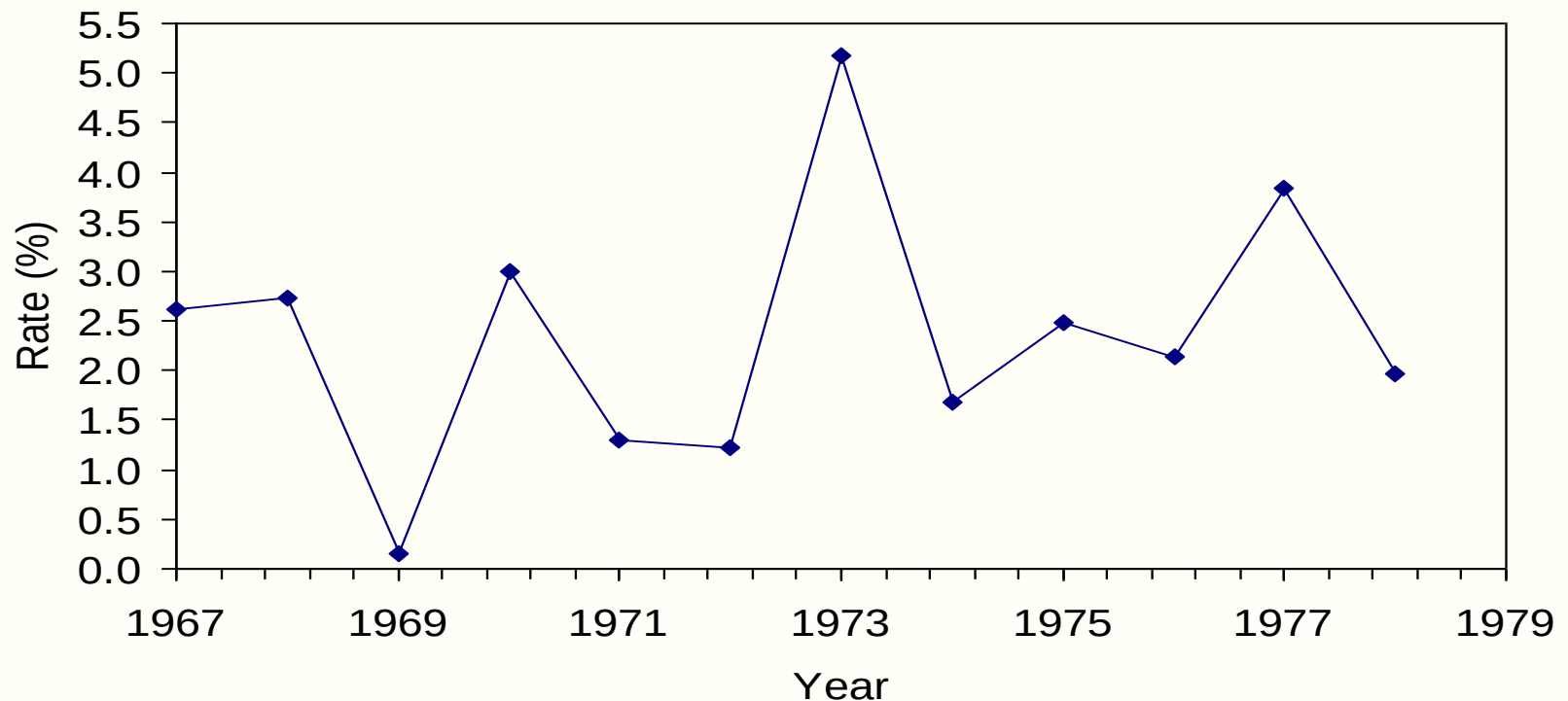


Fig 5: Malaria Parasite Prevalence Rates in Ethiopia, 1967 – 1979 Eth. C.

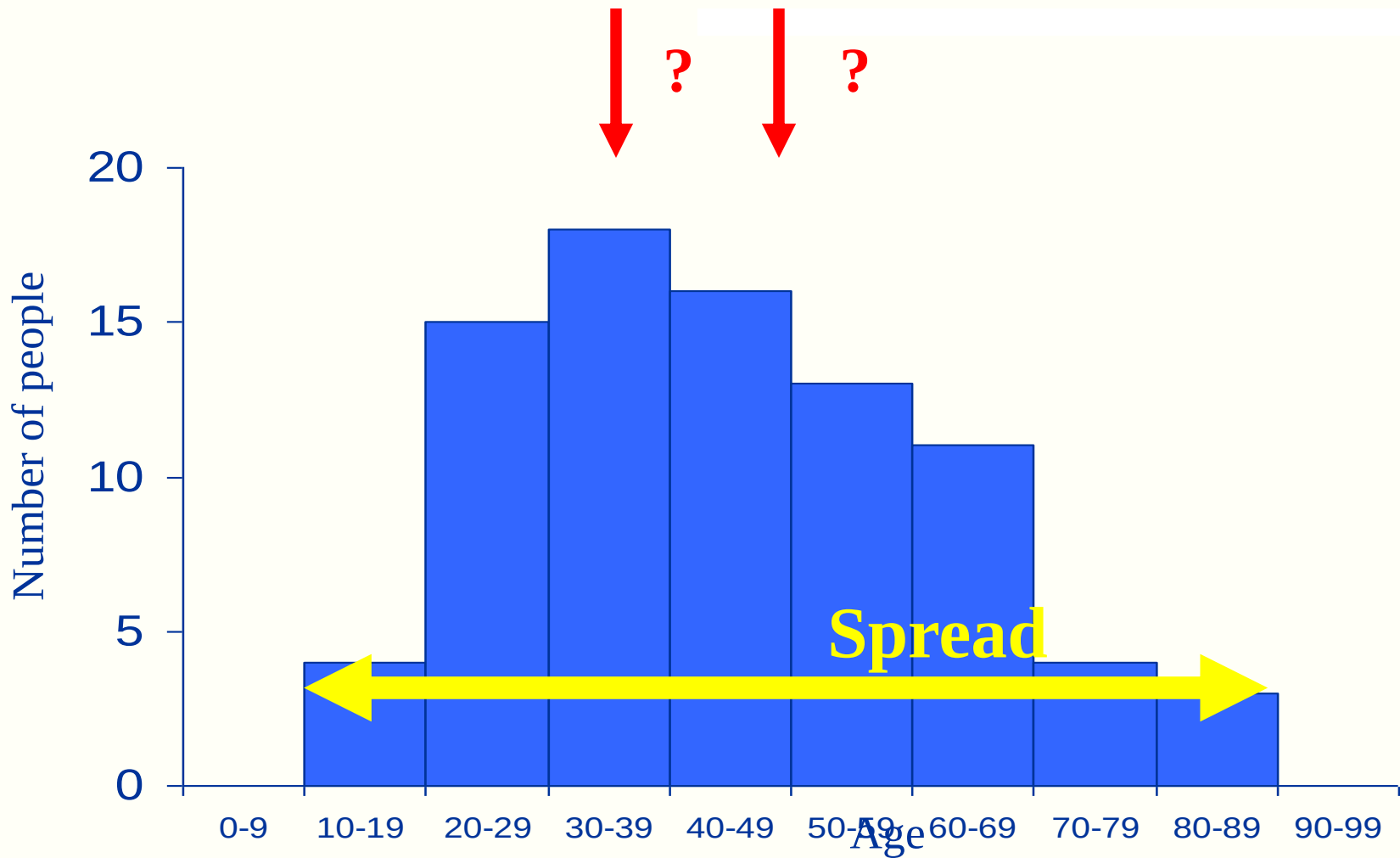
Describing Quantitative Variables

- Measures of Central Location
 - Mean, Median, Mode
- Measures of Spread
 - Range, IQR, Variance, Standard deviation

Measure of Central Location

- **Central Location / Position / Tendency** –
 - A single value that represents (is a good summary of) an entire distribution of data
- Also known as:
 - “Measure of central tendency”
 - “Measure of central position”
- Common measures
 - Arithmetic mean
 - Median
 - Mode

Central Location



	Age
	27
	30
	28
	31
	28
	36
	29
	37
	29
	34
	30
	30
	27
	30
	28
	31
	32
	30
	29
	29

Raw data set:
Ages of students in a class (years)

O bs	Age
1	27
2	27
3	28
4	28
5	28
6	29
7	29
8	29
9	29
10	30
11	30
12	30
13	30
14	30
15	31
16	31
17	32
18	34
19	36
20	37

Order the data set from the lowest
value to the highest value

Add observation numbers

Mode

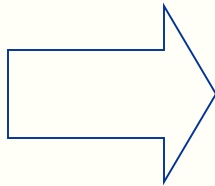
Definition: Mode is the value that occurs most frequently

Method for identification

1. Arrange data into frequency distribution or histogram, showing the values of the variable and the frequency with which each value occurs
2. Identify the value that occurs most often

Mode

Ob s	Age
1	27
2	27
3	28
4	28
5	28
6	29
7	29
8	29
9	29
10	30
11	30
12	30
13	30
14	30
15	31
16	31
17	32
18	34
19	36
20	37



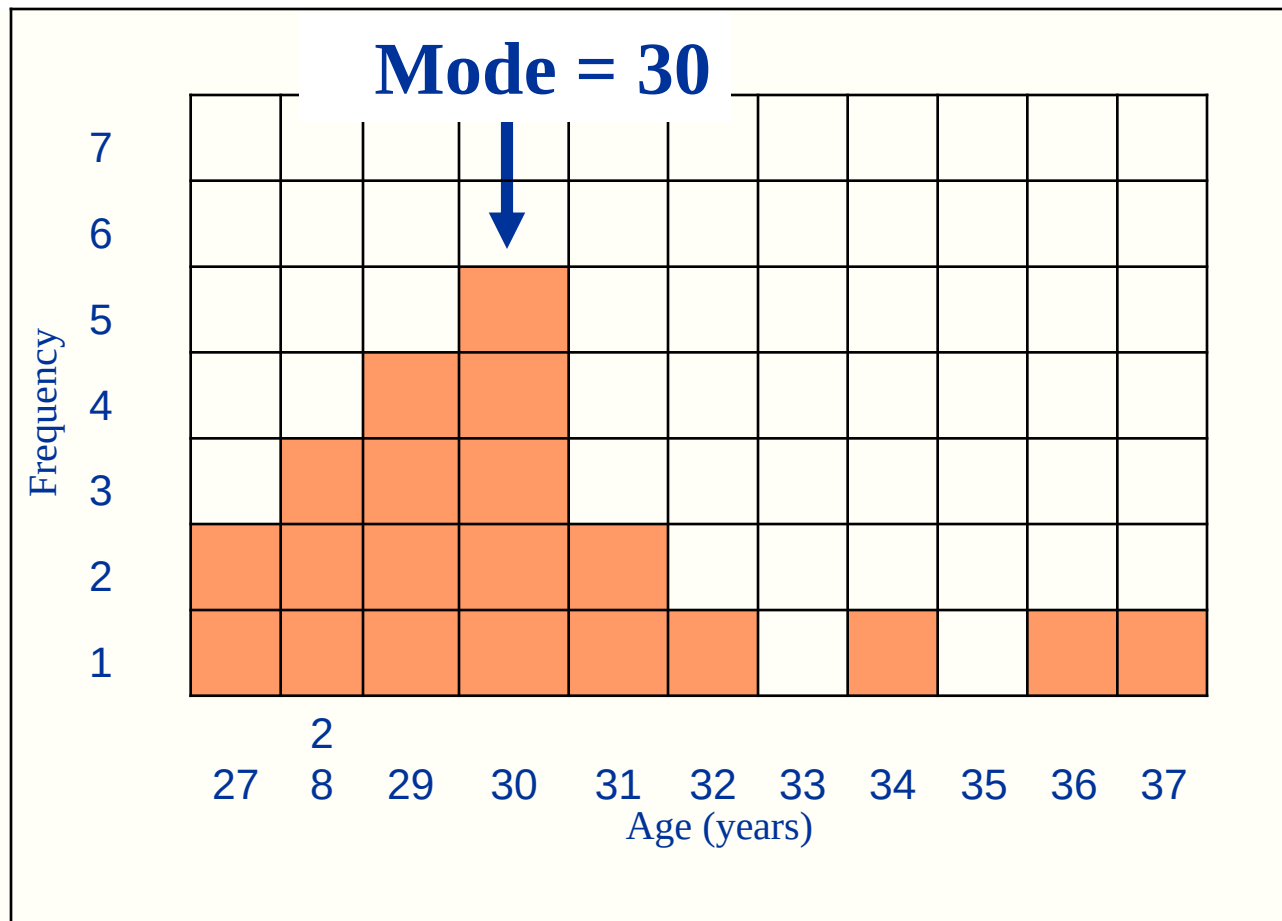
Age	Frequency
27	2
28	3
29	4
30	5
31	2
32	1
33	0
34	1
35	0
36	1
37	1
Total	20



Mode

Mode

The most frequent value of the variable



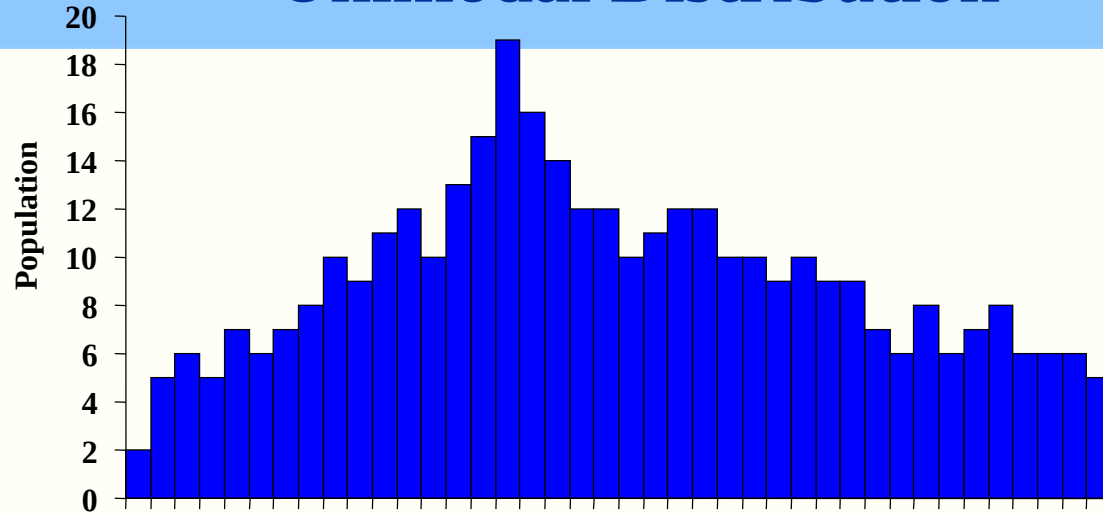
Ob s	Age
1	27
2	27
3	28
4	28
5	28
6	29
7	29
8	29
9	29
10	30
11	30
12	30
13	30
14	30
15	31

Example

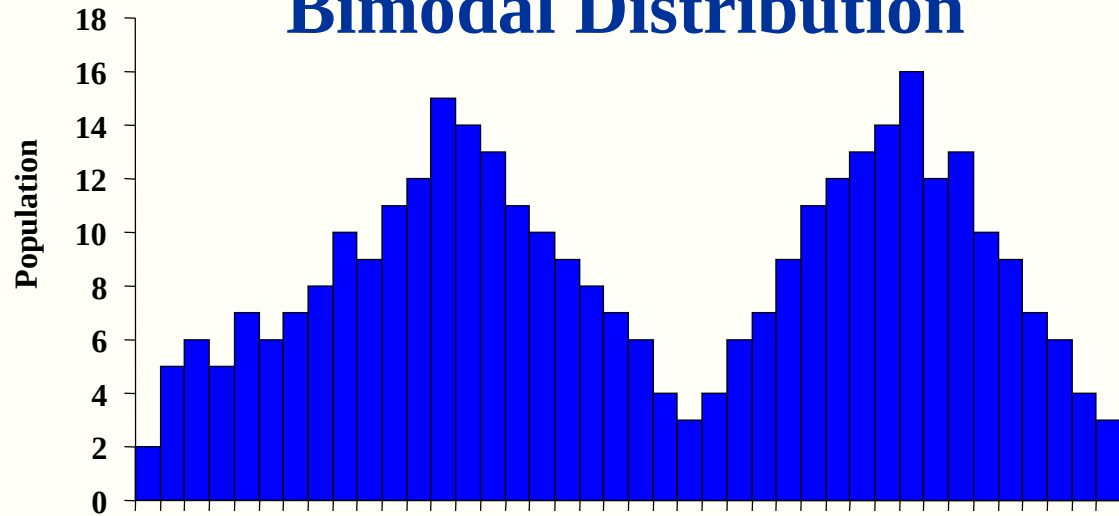
Finding Mode from Length of Stay Data

0, 2, 3, 4, 5, 5, 6, 7, 8, 9,
9, 9, 10, 10, 10, 10, 10, 11, 12, 12,
12, 13, 14, 16, 18, 18, 19, 22, 27, 49

Unimodal Distribution



Bimodal Distribution

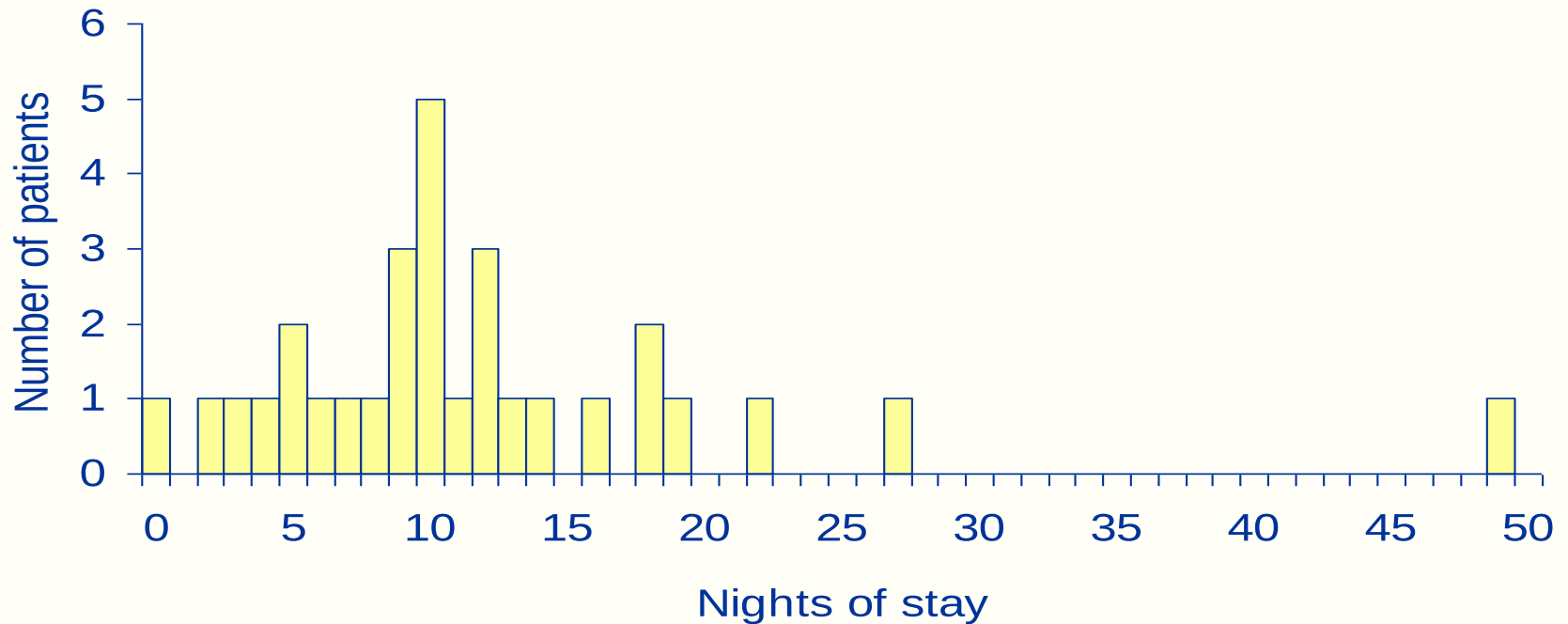


HOSPDAYS	Freq	Percent	Cum.
0	1	3.3%	3.3%
2	1	3.3%	6.7%
3	1	3.3%	10.0%
4	1	3.3%	13.3%
5	2	6.7%	20.0%
6	1	3.3%	23.3%
7	1	3.3%	26.7%
8	1	3.3%	30.0%
9	3	10.0%	40.0%
10	5	16.7%	56.7%
11	1	3.3%	60.0%
12	3	10.0%	70.0%
13	1	3.3%	73.3%
14	1	3.3%	76.7%
16	1	3.3%	80.0%
18	2	6.7%	86.7%
19	1	3.3%	90.0%
22	1	3.3%	93.3%
27	1	3.3%	96.7%
49	1	3.3%	100.0%
Total	30	100.0%	

Mode = 10



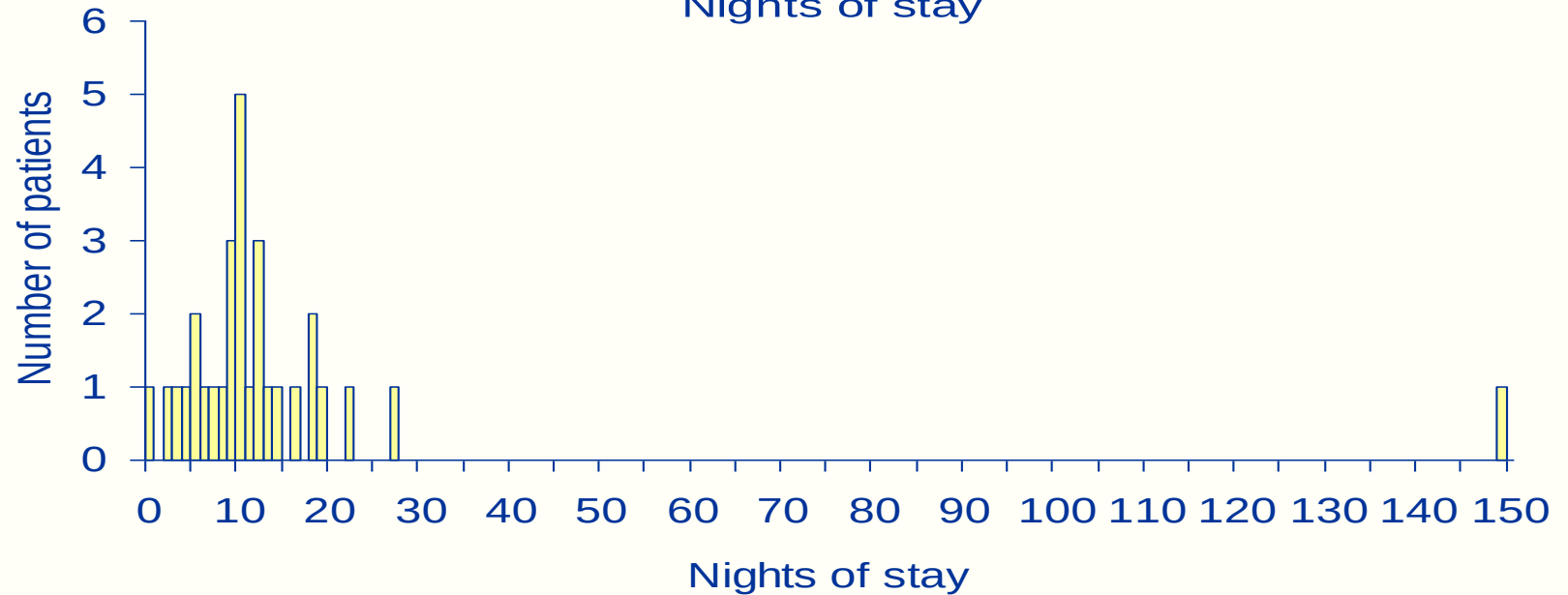
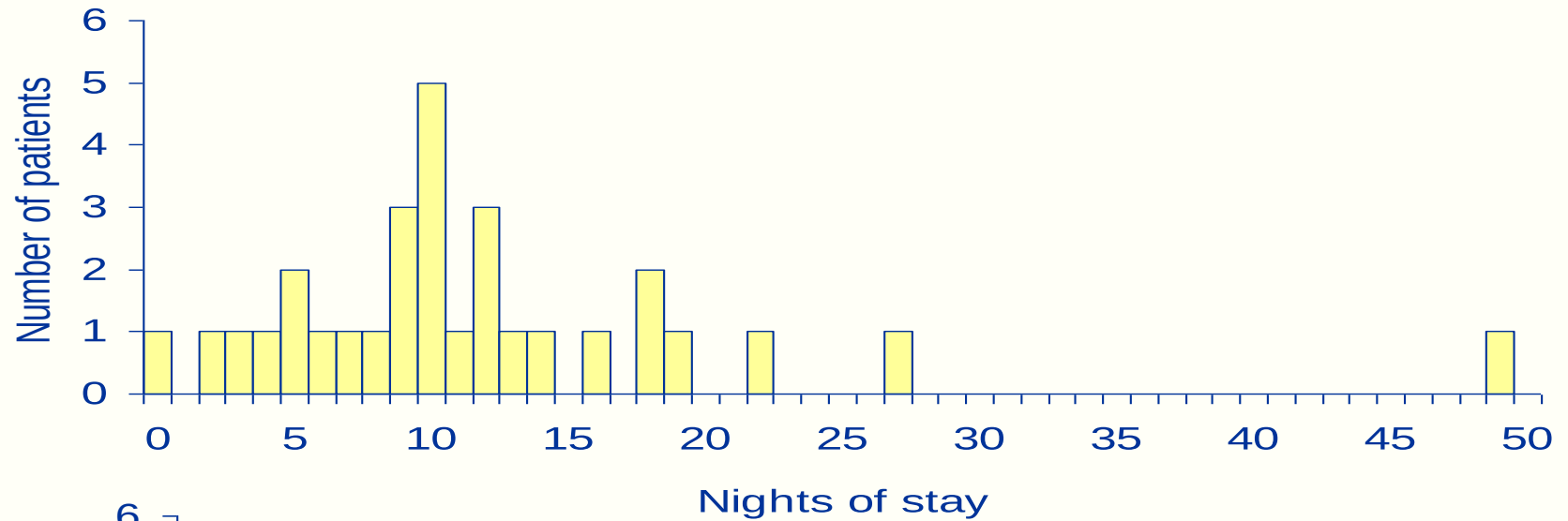
Finding Mode from Histogram



Mode – Properties / Uses

- ◆ Easiest measure to understand, explain, identify
- ◆ Always equals an original value
- ◆ Insensitive to extreme values (outliers)
- ◆ Good descriptive measure, but poor statistical properties
- ◆ May be more than one mode
- ◆ May be no mode
- ◆ Does not use all the data

Outliers



Mode for Grouped data

If data are given in the shape of continuous frequency distribution, the mode is defined as:

$$\hat{X} = L_{mo} + \frac{(\Delta_1)w}{(\Delta_1 + \Delta_2)}$$

Where:

$\hat{X}$ = mode of the distribution.

L_{mo} = the lower class boundary of the modal class.

W = the width of the modal class.

$\Delta_1 = f_{mo} - f_1$; $\Delta_2 = f_{mo} - f_2$

f_{mo} is frequency of the modal class.

f_1 is the frequency immediately preceding the modal class.

f_2 is the frequency immediately succeeding the modal class.

Example

Calculate the mode of the distribution.

Age	5- 15	15- 25	25- 35	35- 45	45- 55	55- 65	65- 75
Frequenc	8	12	17	29	31	5	3

Solutions:

45-55 is the modal classes since it contain the highest frequency.

$$L_{mo}=45$$

$$f_{mo}=31$$

$$W=10$$

$$\Delta_1 = f_{mo} - f_1 = 31 - 29 = 2$$

$$\Delta_2 = f_{mo} - f_2 = 31 - 5 = 26$$

Then apply the formula for mode

$$\hat{X} = L_{mo} + \frac{(\Delta_1)W}{(\Delta_1 + \Delta_2)} = 45 + \left(\frac{2}{2 + 26} * 10 \right) = 45.71$$

Median

Definition: Median is the middle value; also, the value that splits the distribution into two equal parts

- *50% of observations are below the median*
- *50% of observations are above the median*

Method for identification

1. Arrange observations in order
2. Find middle position as $(n + 1) / 2$ or $(n/2)$
3. Identify the value at the middle

Median: Odd Number of Values

Obs	Age
1	27
2	27
3	28
4	28
5	28
6	29
7	29
8	29
9	29
10	30
11	30
12	30
13	30
14	30
15	31
16	31
17	32
18	34
19	36



N = 19

$$\begin{aligned}\text{Median Observation} &= \frac{N+1}{2} \\ &= \frac{19+1}{2} \\ &= \frac{20}{2} \\ &= 10\end{aligned}$$

Median age = 30 years

Obs	Age
1	27
2	27
3	28
4	28
5	28
6	29
7	29
8	29
9	29
10	30
11	30
12	30
13	30
14	30
15	31
16	31
17	32
18	34
19	36
20	37

Median: Even Number of Values

$$N = 20$$

$$\begin{aligned} \text{Median Observation} &= \frac{N+1}{2} \\ &= \frac{20+1}{2} \\ &= \frac{21}{2} \\ &= 10.5 \end{aligned}$$



Median age = Average value between
10th and 11th observation

$$\frac{30+30}{2} = 30 \text{ years}$$

Examples

Find Median of Length of Stay Data;

0, 2, 3, 4, 5, 5, 6, 7, 8, 9,
9, 9, 10, 10, 10, 10, 10, 11, 12, 12,
12, 13, 14, 16, 18, 18, 19, 22, 27, 49

HOSPDAYS	Freq	Percent	Cum.
0	1	3.3%	3.3%
2	1	3.3%	6.7%
3	1	3.3%	10.0%
4	1	3.3%	13.3%
5	2	6.7%	20.0%
6	1	3.3%	23.3%
7	1	3.3%	26.7%
8	1	3.3%	30.0%
9	3	10.0%	40.0%
10	5	16.7%	56.7%
11	1	3.3%	60.0%
12	3	10.0%	70.0%
13	1	3.3%	73.3%
14	1	3.3%	76.7%
16	1	3.3%	80.0%
18	2	6.7%	86.7%
19	1	3.3%	90.0%
22	1	3.3%	93.3%
27	1	3.3%	96.7%
49	1	3.3%	100.0%
Total	30	100.0%	

Median at 50%
= 10

Median for grouped data

- ◆ -If data are given in the shape of continuous frequency distribution, the median is defined as:

- ◆
$$\tilde{X} = L_{\text{med}} + \frac{W}{f_{\text{med}}} \left(\frac{n}{2} - C_{\text{fp}} \right)$$

Where:

L_{med} = the lower class boundary of the median class.

W = the width of the median class.

n = the total number of observation.

C_{fp} = less than cumulative frequency of the class immediately preceding the median class.

f_{med} = frequency of the median class.

Example

Find the median of the following age distribution.

Class interval	40-44	45-49	50-54	55-59	60-64	65-69	70-74
Frequency	7	10	22	15	12	6	3

Solutions:

- First find the less than cumulative frequency.
- Identify the median class by dividing n by 2.
- Find median using the formula.

Class interval	40-44	45-49	50-54	55-59	60-64	65-69	70-74
Frequency	7	10	22	15	12	6	3
CF	7	17	39	54	66	72	75

Cont...

- ◆ The median class is $n/2=75/2=37.5$
- ◆ Since the 37.5th observation lie on the third class, therefore, the median class will be on the third class or the first cumulative frequency (less than type) greater than or equal to 37.5 is 39.
- ◆
$$\tilde{X} = L_{\text{med}} + \frac{W}{f_{\text{med}}} \left(\frac{n}{2} - \text{Cfp} \right)$$
- ◆
$$= 49.5 + \frac{5}{22} \left(\frac{75}{2} - 17 \right) = 54.16$$
- ◆ Therefore, the mid-point (the median age) of the distribution will be 54.16.

Median – Properties / Uses

- ◆ Does not use all the data available
- ◆ Insensitive to extreme values (outliers)
- ◆ Good descriptive measure but poor statistical properties
- ◆ Measure of choice for skewed data
- ◆ Equals an original value of n is odd

Arithmetic Mean

Arithmetic mean = “average” value

Method for identification

1. Sum up all of the values
2. Divide the sum by the number of observations (n)

Arithmetic Mean

Obs	Age
1	27
2	27
3	28
4	28
5	28
6	29
7	29
8	29
9	29
10	30
11	30
12	30
13	30
14	30
15	31
16	31
17	32
18	34
19	36
20	37

$$\bar{\mathbf{m}} = \frac{\sum \mathbf{x}_i}{\mathbf{N}}$$

$$\mathbf{N} = 20$$

$$\sum \mathbf{x}_i = 605$$

$$\bar{\mathbf{m}} = \frac{605}{20} = 30.25$$

Example

Finding the Mean — Length of Stay Data

0, 2, 3, 4, 5, 5, 6, 7, 8, 9,

9, 9, 10, 10, 10, 10, 10, 11, 12, 12,

12, 13, 14, 16, 18, 18, 19, 22, 27, 49

$$\text{Sum} = 360$$

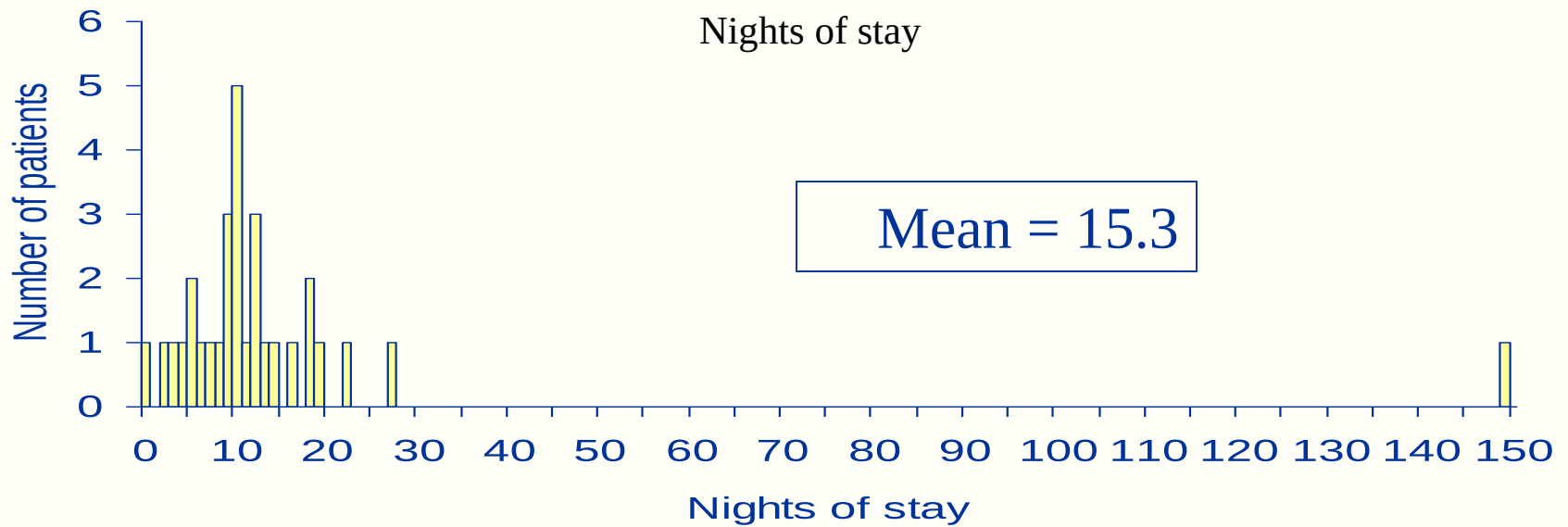
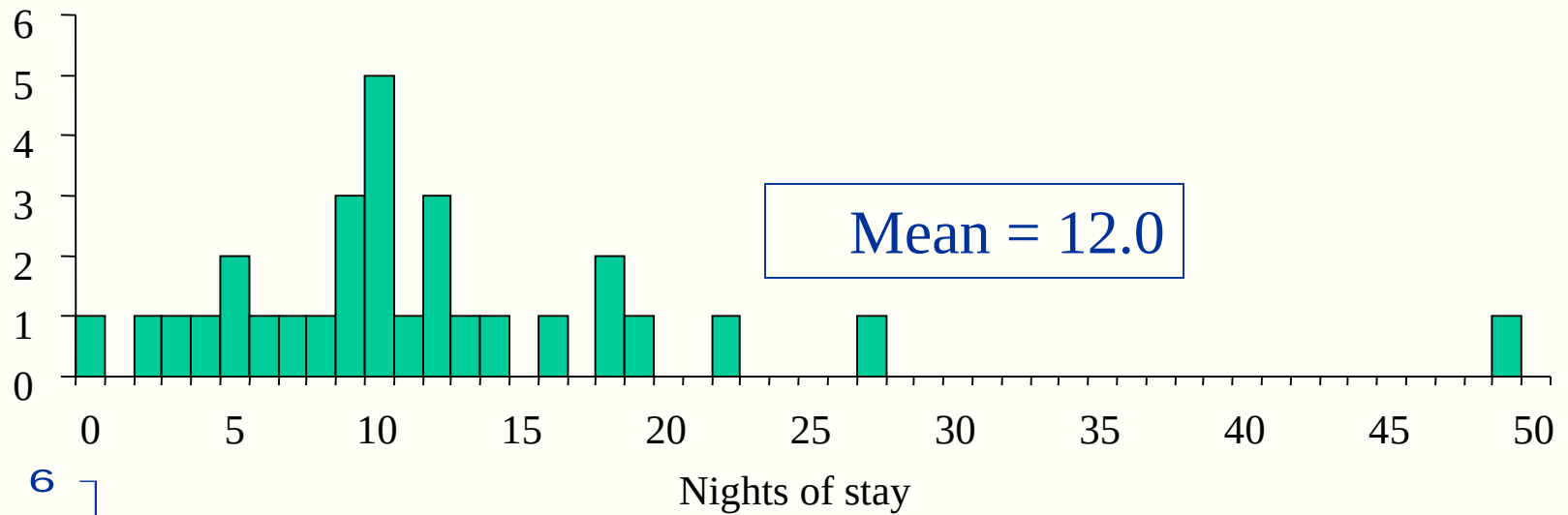
$$n = 30$$

$$\text{Mean} = 360 / 30 = ?$$

Arithmetic Mean – Properties

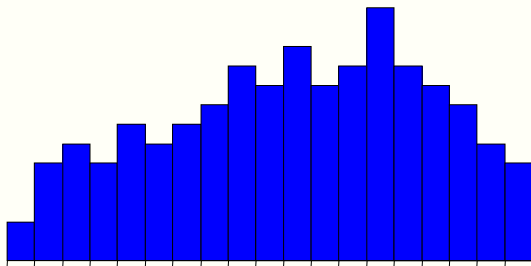
- ◆ Probably best known measure of central location
- ◆ Use all of the data
- ◆ Affected by extreme values (outliers)
- ◆ Best for normally distributed data
- ◆ Not usually equal to one of the original values
- ◆ Good statistical properties

Sensitive to Outliers

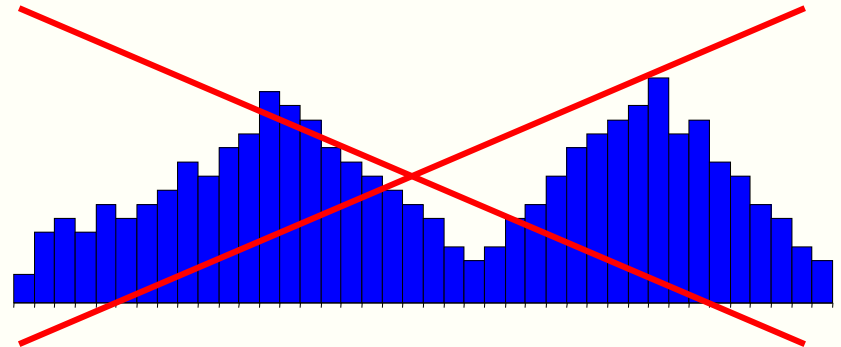
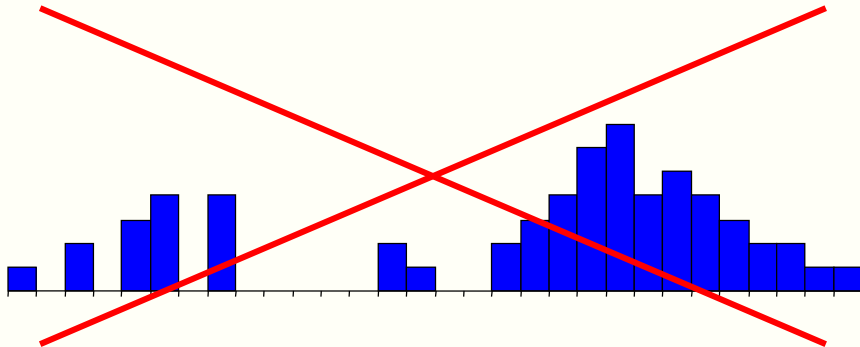
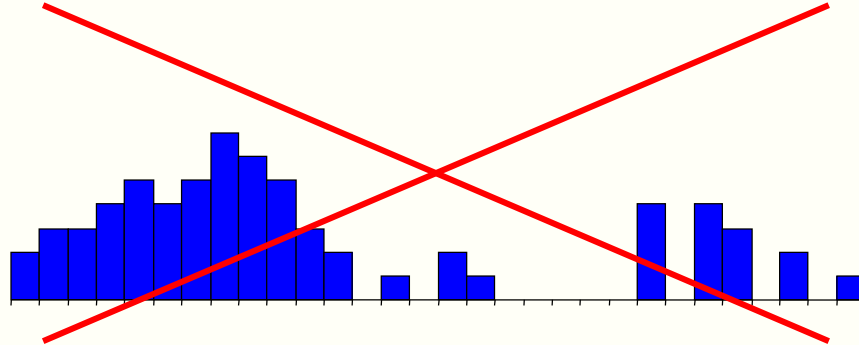


When to use the arithmetic mean?

- Centered distribution
- Approximately symmetrical
- Few extreme values (outliers)



OK!



Arithmetic Mean for Grouped Frequency Distribution

If data are given in the form of **continuous frequency distribution**, the sample mean can be computed as

$$\bar{X} = \frac{f_1x_1 + f_2x_2 + \dots + f_kx_k}{f_1 + f_2 + \dots + f_k} = \frac{\sum_{i=1}^k f_i x_i}{\sum_{i=1}^k f_i}$$

Where x_i is **the class mark** of the i^{th} class;

f_i the frequency of the i^{th} class

k = the number of classes

Arithmetic Mean for Grouped Frequency Distribution

Example 1: calculate the mean (AM) for the following sample data.

Class limit	Class Mark	Frequency
6 – 11	8.5	2
12 – 17	14.5	2
18 – 23	20.5	7
24 – 29	26.5	4
30 – 35	32.5	3
36 – 41	38.5	2

$$\bar{X} = \frac{f_1x_1 + f_2x_2 + \dots + f_kx_k}{f_1 + f_2 + \dots + f_k} = \frac{\sum_{i=1}^k f_i x_i}{\sum_{i=1}^k f_i} = \frac{2(8.5) + 2(14.5) + \dots + 2(38.5)}{2 + 2 + \dots + 2} = 23.5$$

The weighted mean

- ◆ In some cases the data in the sample or population should not be weighted equally, and each value weighted according to its importance. Weighted arithmetic mean is used to calculate the average when the relative importance of the observations differs. This relative importance is technically known as weight. Weight could be a frequency or numerical coefficient associated with observations.
- ◆ **Definition:** If $x_1, x_2, x_3 \dots x_n$ have weights $w_1, w_2, w_3 \dots w_n$ respectively, then the weighted mean $\bar{X}_w$, is defined as:
- ◆
$$\bar{X}_w = \frac{x_1w_1 + x_2w_2 + x_3w_3 + \dots + x_nw_n}{w_1 + w_2 + w_3 + \dots + w_n} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}$$

Example:

The GPA or CGPA of a student is a good example of a weighted arithmetic mean. Suppose that a student obtained the following grades in the first semester of the freshman program at Addis Ababa University in 2009. Find the GPA of a student.

Course	Credit hour (W_i)	Grade (X_i)
Math 101	4	A=4
Bio 101	3	C=2
Stat 101	3	B=3
Phys 101	4	B=3
Flen 101	3	C=2

$$\bar{X}_w = \text{GPA} = \frac{(4 \times 4) + 3 \times 2 + 3 \times 3 + 4 \times 3 + 3 \times 2}{4 + 3 + 3 + 4 + 3} = \frac{49}{17} = 2.88$$

The Grand mean

- ◆ If $\bar{X}_1$ is the mean of observations n_1 ,
- ◆ If $\bar{X}_2$ is the mean of observations $n_2, \dots$,
If $\bar{X}_k$ is the mean of observations n_k
then, the mean of all the observation
in all groups often called the combined
mean (Grand mean) is given by:

$$\bar{X}_c = \frac{\bar{X}_1 n_1 + \bar{X}_2 n_2 + \bar{X}_3 n_3 + \dots + \bar{X}_k n_k}{n_1 + n_2 + n_3 + \dots + n_k} = \frac{\sum_{i=1}^k n_i \bar{X}_i}{\sum_{i=1}^k n_i}$$

- ◆ Example: In a class there are 30 females and 70 males. If females averaged 60 in an examination and boys averaged 72, find the mean for the entire class.

- ◆ Solution:

- ◆
$$\bar{X}_C = \frac{\sum_{i=1}^2 ni\bar{X}_i}{\sum_{i=1}^2 ni} = \frac{30(60)+70(72)}{30+70} = 68.40$$

Correct mean

- ◆ If a wrong figure has been used when calculating the mean the correct mean can be obtained without repeating the whole process using:

$$\text{Correct mean} = \text{Wrong mean} + \frac{(\text{Correct value} - \text{Wrong value})}{n}$$

- ◆ Example: An average weight of 10 patients was calculated to be 65. Later it was discovered that one weight was misread as 40 instead of 80 k.g. Calculate the correct average weight.
- ◆ solution

$$\text{Correct mean} = 65 + \frac{(80 - 40)}{10} = 69 \text{ kg}$$

The effect of transforming original series on the mean.

- ◆ If a constant k is added/ subtracted to/from every observation then the new mean will be the old mean $\pm k$ respectively.
- ◆ If every observations are multiplied by a constant k then the new mean will be $k \times \text{old mean}$

Quartiles

Definition: Quartile is the value that splits the distribution into four equal parts

- *25% of observations are below the first quartile (Q1)*
- *25% of observations are between Q1 and Q2 (median)*
- *25% of observations are between Q2 (median) and Q3*
- *25% of observations are above Q3*

Quartiles

	Obs	Age
	1	27
	2	27
	3	28
	4	28
Q1	5	28
	6	29
	7	29
	8	29
	9	29
Q2	10	30
	11	30
	12	30
	13	30
	14	30
	15	31
Q3	16	31
	17	32
	18	34
	19	36
	20	37

Q1 age = 28 Q2 age = 30 Q3 age = 31

$$\begin{aligned}
 \text{Q1 observation} &= \text{round}\left\{\frac{N+1}{4}\right\} \\
 &= \frac{20+1}{4} = \frac{21}{4} \\
 &= 5.25 \sim 5^{\text{th}} \text{ obs}
 \end{aligned}$$

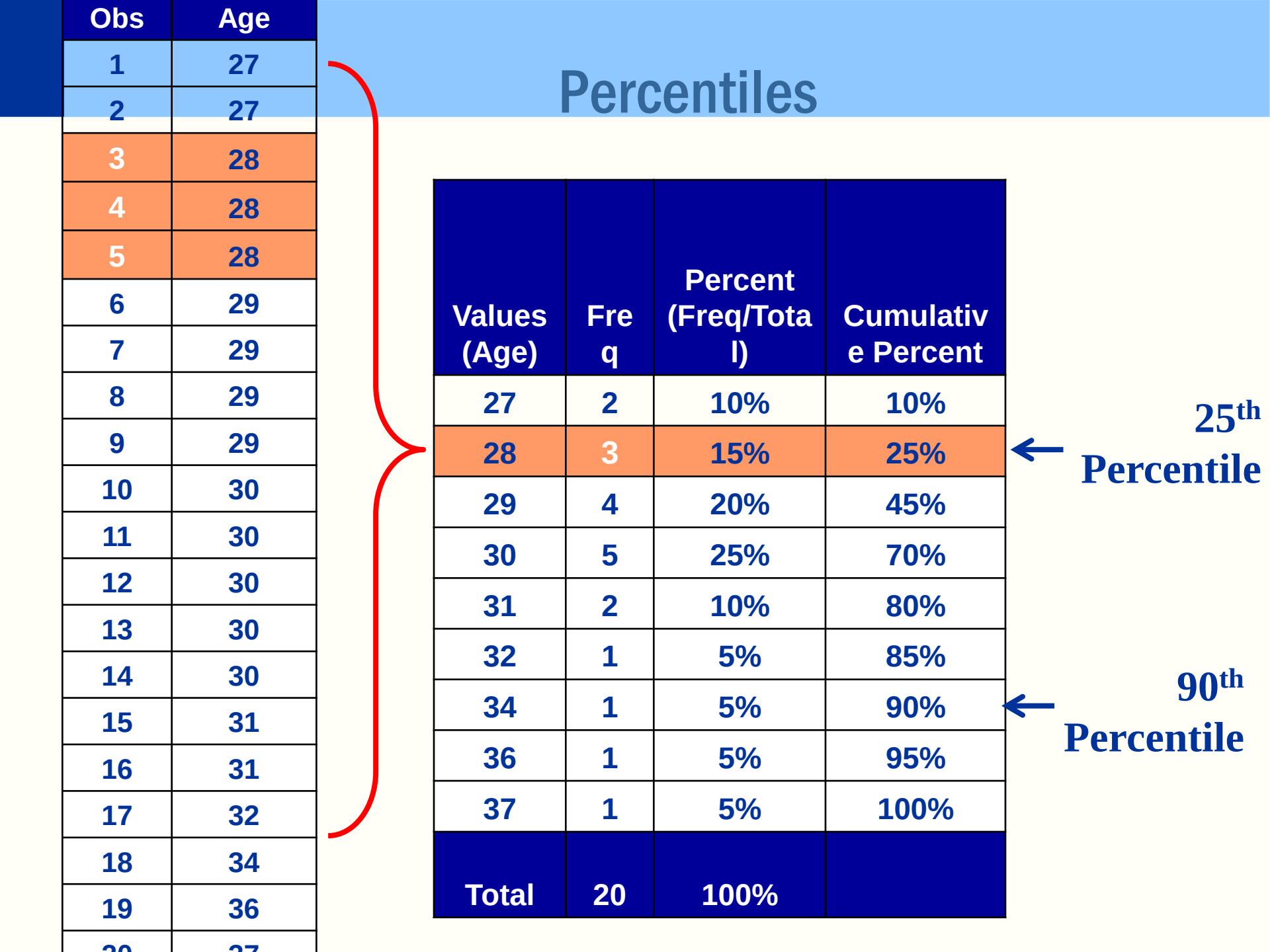
Q2 observation = 10.5 (median)

$$\begin{aligned}
 \text{Q3 observation} &= \text{round}\left\{\frac{3(N+1)}{4}\right\} \\
 &= \frac{3(20+1)}{4} = \frac{3(21)}{4} \\
 &= 15.75 \sim 16^{\text{th}} \text{ obs}
 \end{aligned}$$

Percentiles

Value of the variable that splits the distribution in 100 equal parts

- *35 % of observations are below the 35th percentile*
- *65 % of observations are above 35th percentile*



Summary

- Measure of Central Location – single measure that represents an entire distribution
- Mode – most common value
- Median – central value
- Arithmetic mean – average value
- Mean uses all data, so sensitive to outliers
- Mean has best statistical properties
- Mean preferred for normally distributed data
- Median preferred for skewed data
- Geometric mean for dilutional titer

Measures of Spread

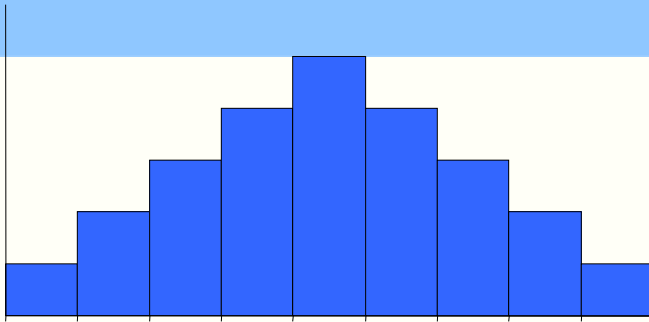
Definition: Measures that quantify the variation or dispersion of a set of data from its central location

Also known as:

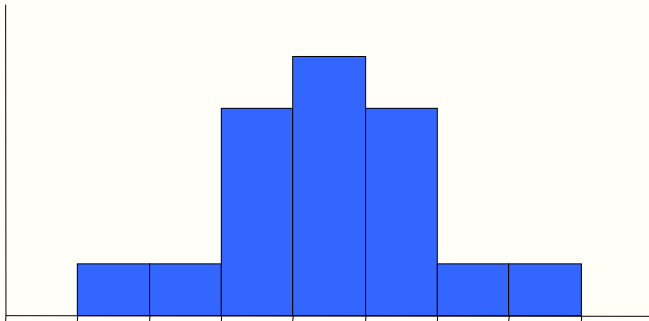
- “Measure of dispersion”
- “Measure of variation”

Common measures

- Range
- Interquartile range
- Variance
- standard deviation

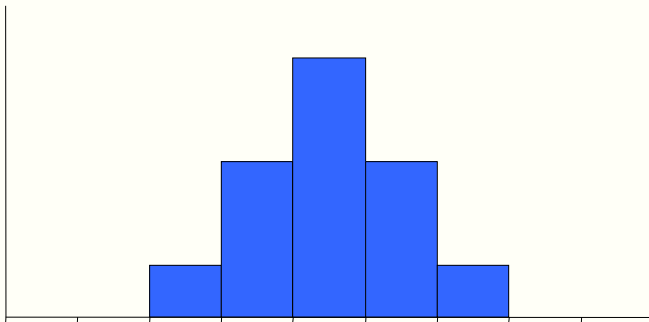


Same center



but ...

different dispersions



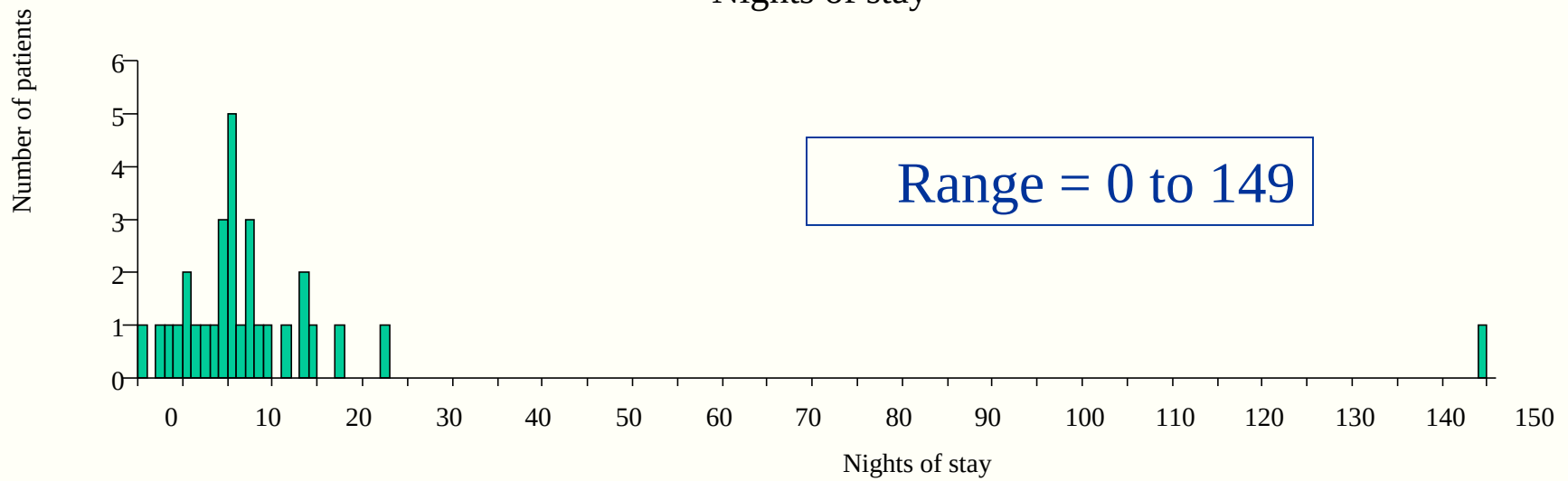
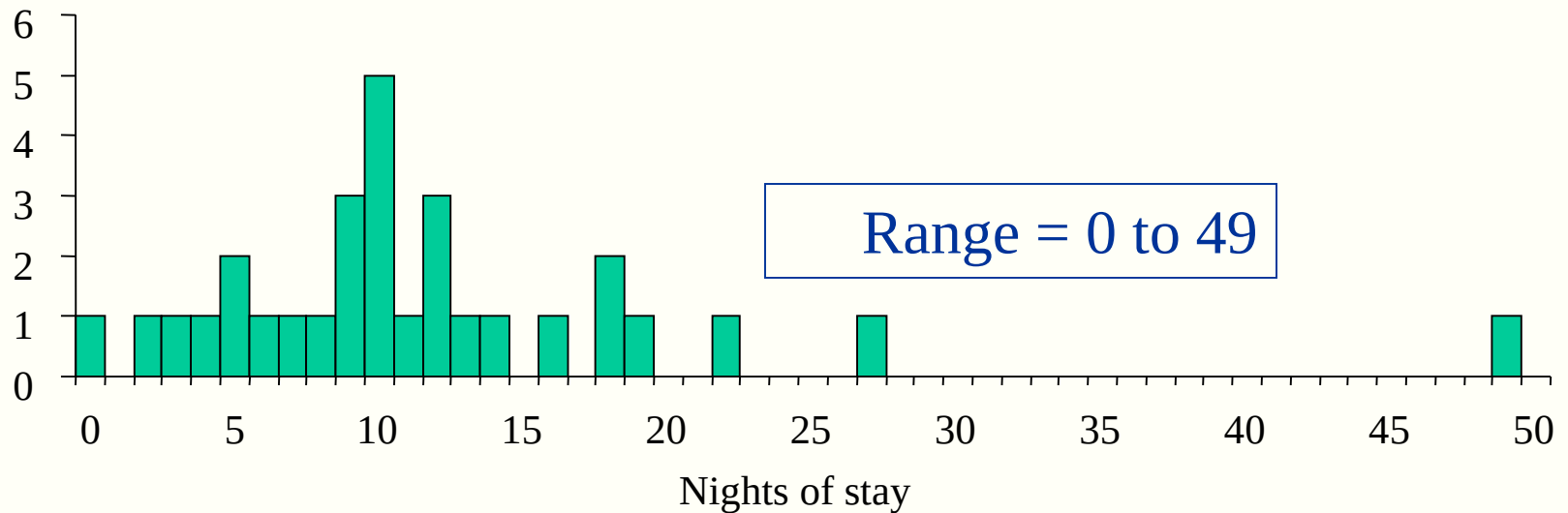
Range

Definition: difference between largest and smallest values

Example: Finding the Range of Length of Stay Data

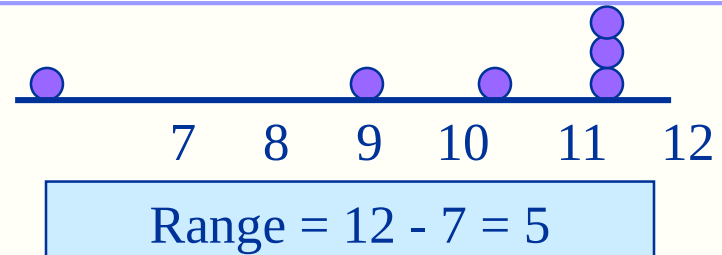
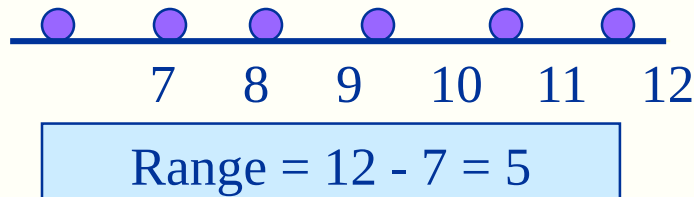
0, 2, 3, 4, 5, 5, 6, 7, 8, 9,
9, 9, 10, 10, 10, 10, 10, 11, 12, 12,
12, 13, 14, 16, 18, 18, 19, 22, 27, 49

Range – Sensitive to Outliers?



Disadvantages of the Range

- ◆ Ignores the way in which data are distributed



- ◆ Sensitive to outliers

1,1,1,1,1,1,1,1,1,1,1,1,2,2,2,2,2,2,2,2,3,3,3,3,4,5

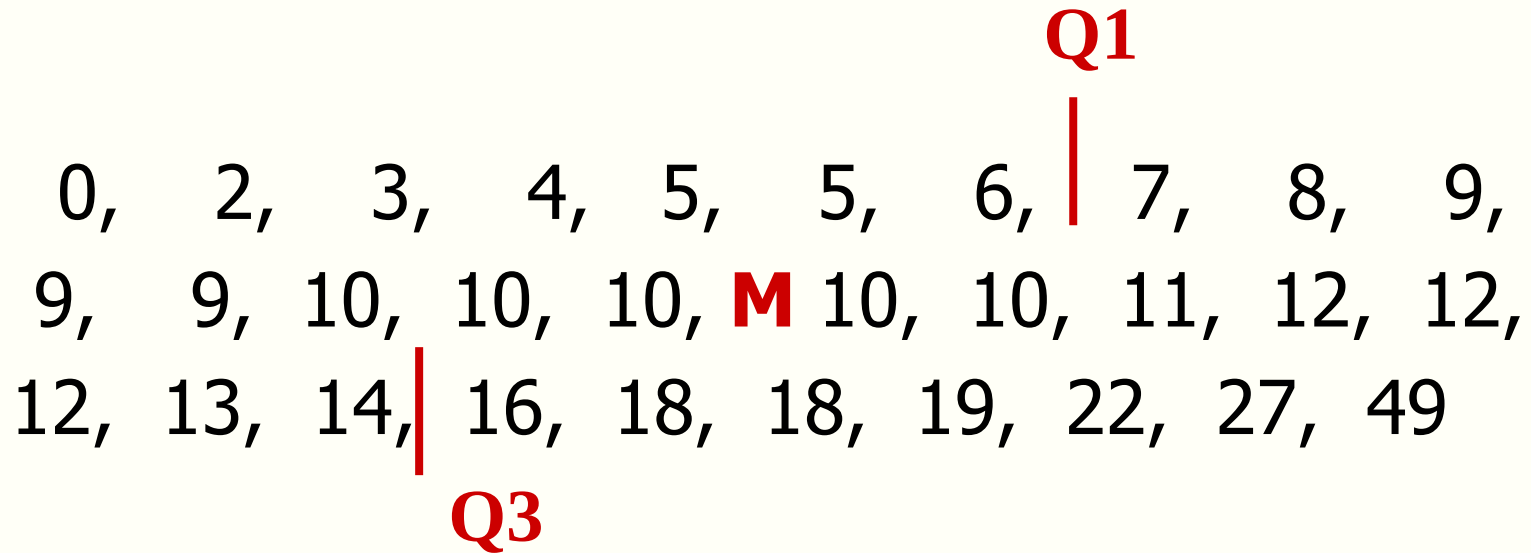
Range = 5 - 1 = 4

1,1,1,1,1,1,1,1,1,1,1,1,1,2,2,2,2,2,2,2,2,3,3,3,3,4,120

Range = 120 - 1 = 119

IQR Example

IQR— Length of Stay Data

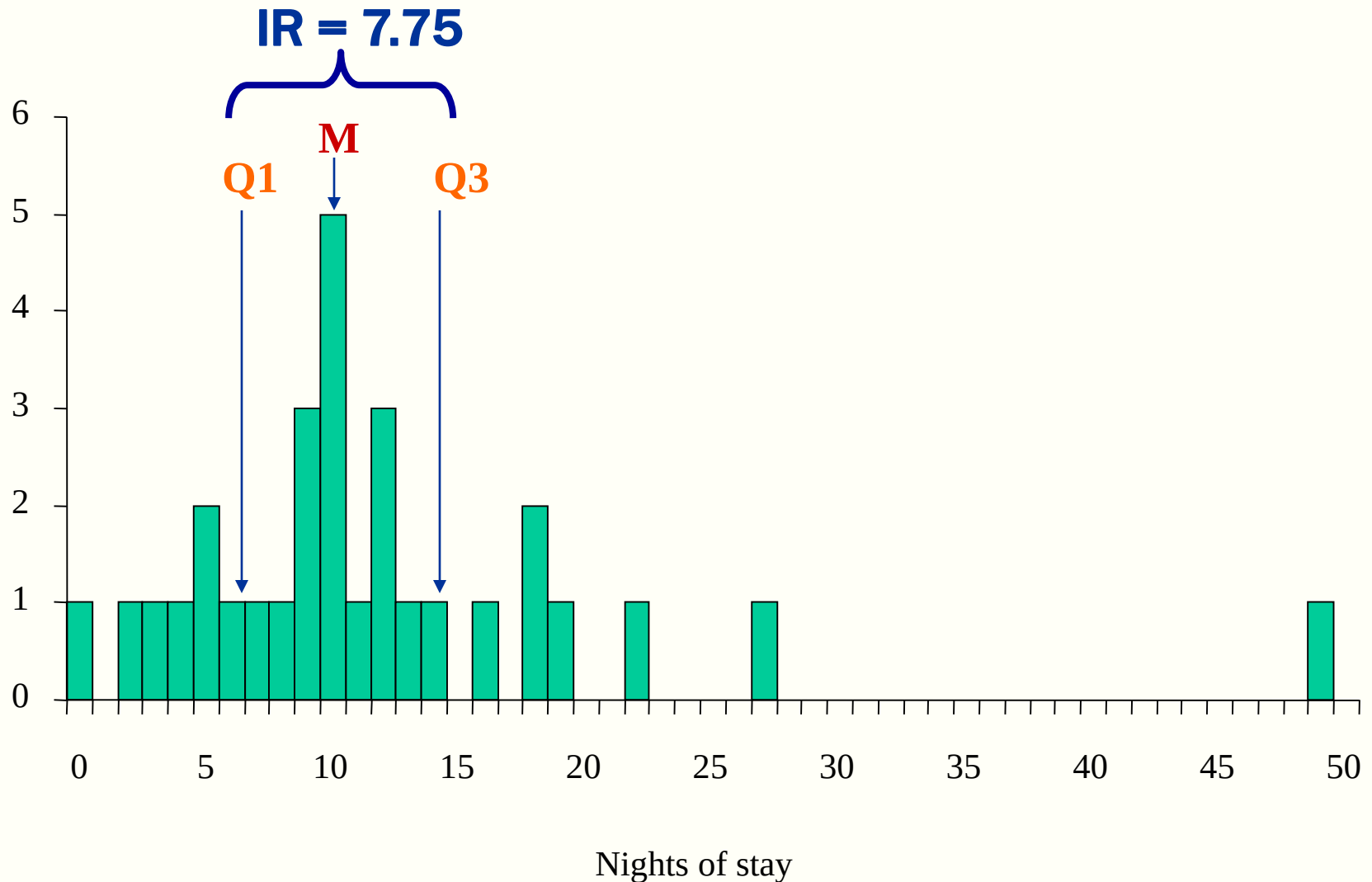


$$\mathbf{Q1} = 25^{\text{th}} \text{ percentile} = (30+1) / 4 = 7\frac{3}{4} \quad \rightarrow 7$$

$$\mathbf{Median} = 50^{\text{th}} \text{ percentile} = 15.5 \quad \rightarrow 10$$

$$\mathbf{Q3} = 75^{\text{th}} \text{ percentile} = 3 (30+1) / 4 = 23\frac{1}{4} \quad \rightarrow 14$$

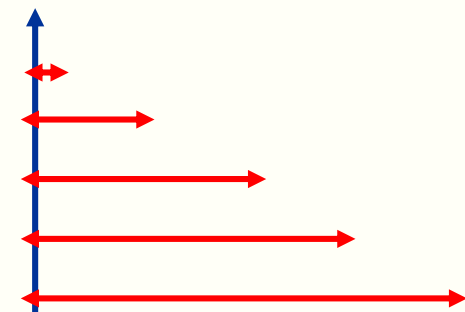
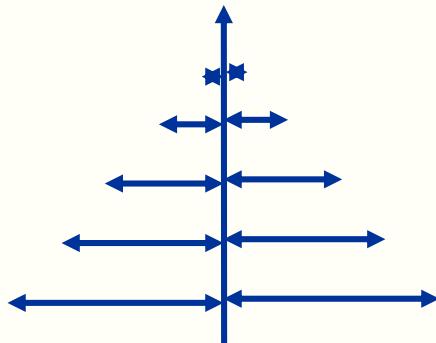
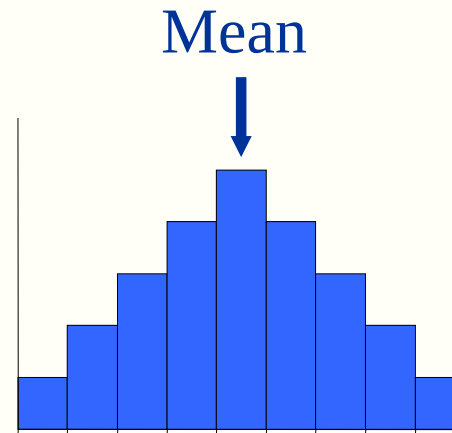
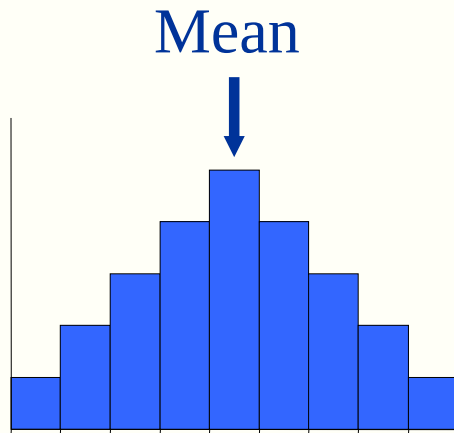
IQR— Length of Stay Data



Sample Variance and sample Standard Deviation

- ◆ Definition: measures of variation that quantifies how closely clustered the observed values are to the mean
- ◆ **Sample Variance** = average of squared deviations from mean
$$= \text{Sum } (x - \text{mean})^2 / n-1$$
- ◆ **Sample Standard deviation** = square root of variance

Variance and Standard Deviation



Equations for sample Variance and sample Standard Deviation

$\bar{x}$: Mean

x_i : Data value

n : No. of observation

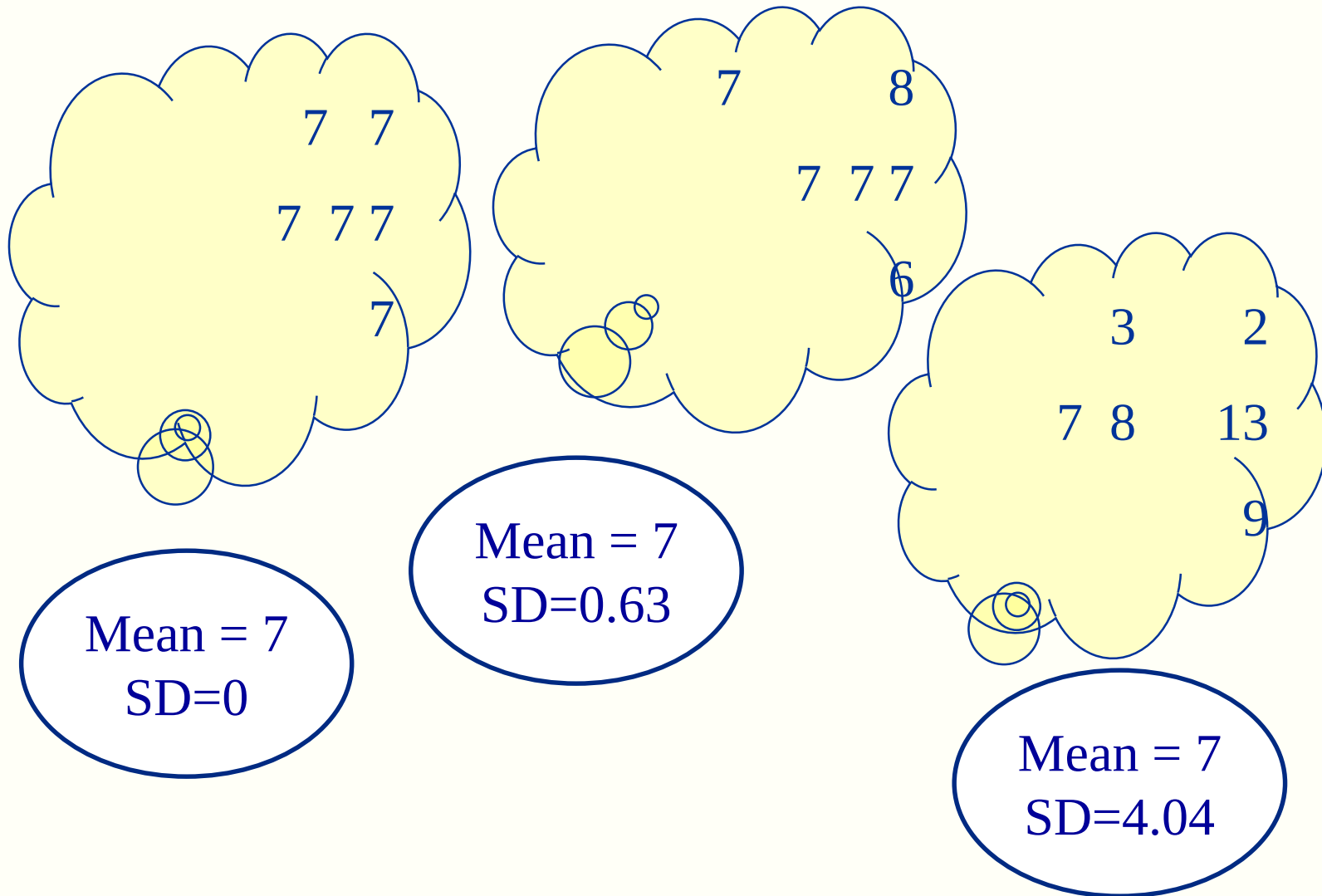
s^2 : Variance

s : Standard deviation

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

Standard deviation SD



Population Variance

- ◆ Average of squared deviations of values from the mean

– Population variance:

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

Where μ = population mean

N = population size

x_i = i^{th} value of the variable x

Population Standard Deviation

- ◆ Most commonly used measure of variation
- ◆ Shows variation about the mean
- ◆ Has the same units as the original data

– Population standard deviation:

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \mu)^2}{N}}$$

Standard Deviation – Properties / Uses

Standard deviation usually calculated only when data are more or less normally distributed (bell shaped curve)

For normally distributed data,

- ◆ 68.3% of the data fall within plus/minus 1 SD
- ◆ 95.5% of the data fall within plus/minus 2 SD
- ◆ 95.0% of the data fall within plus/minus 1.96 SD
- ◆ 99.7% of the data fall within plus/minus 3 SD

Calculation Example: Sample Standard Deviation

Sample

Data (x_i) : 10 12 14 15 17 18 18 24

$$n = 8$$

$$\text{Mean } \bar{x} = 16$$

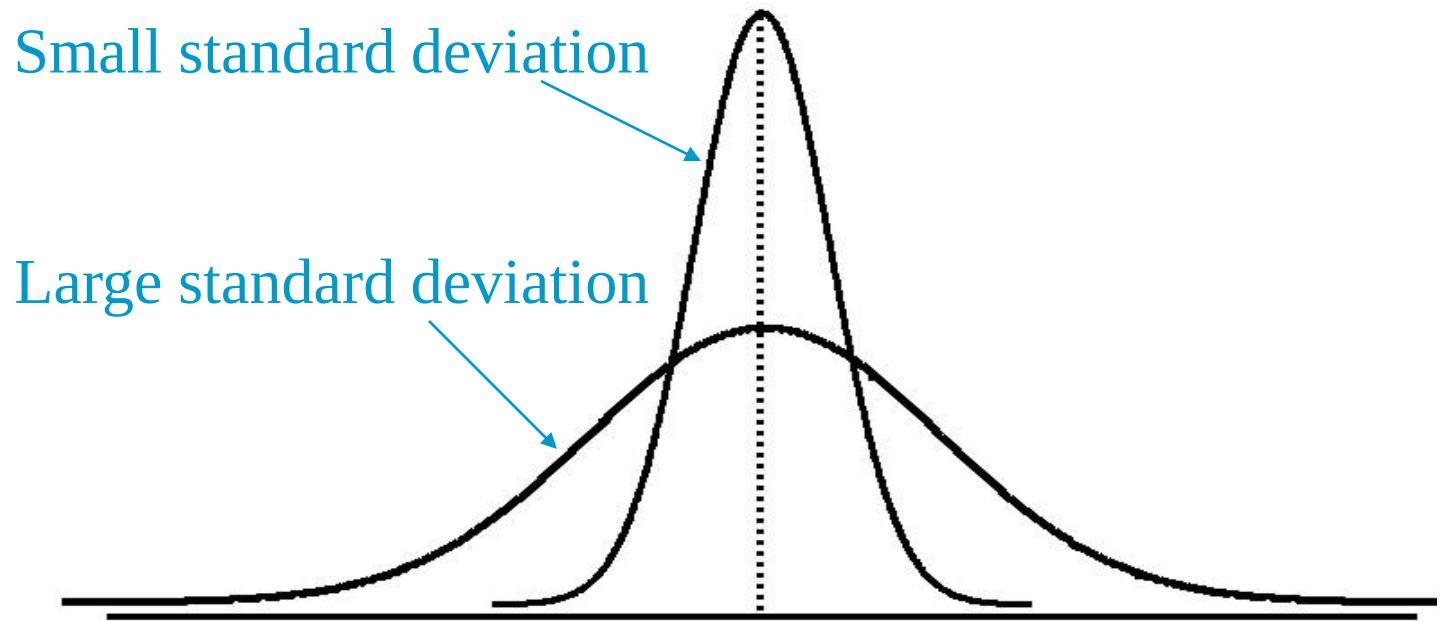
$$s = \sqrt{\frac{(10 - \bar{x})^2 + (12 - \bar{x})^2 + (14 - \bar{x})^2 + \cdots + (24 - \bar{x})^2}{n - 1}}$$

$$= \sqrt{\frac{(10 - 16)^2 + (12 - 16)^2 + (14 - 16)^2 + \cdots + (24 - 16)^2}{8 - 1}}$$

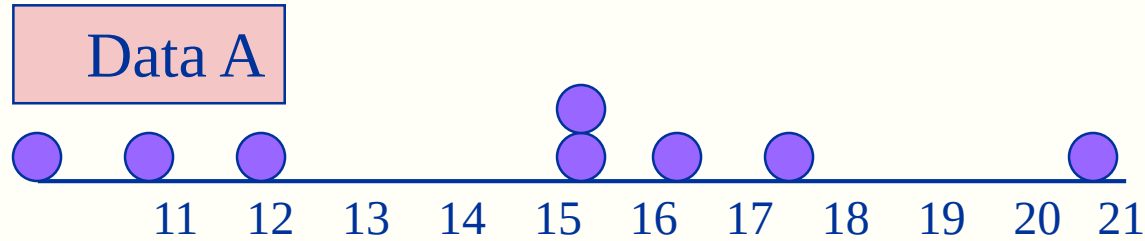
$$= \sqrt{\frac{126}{7}} = 4.2426 \rightarrow$$

A measure of the “average”
scatter around the mean

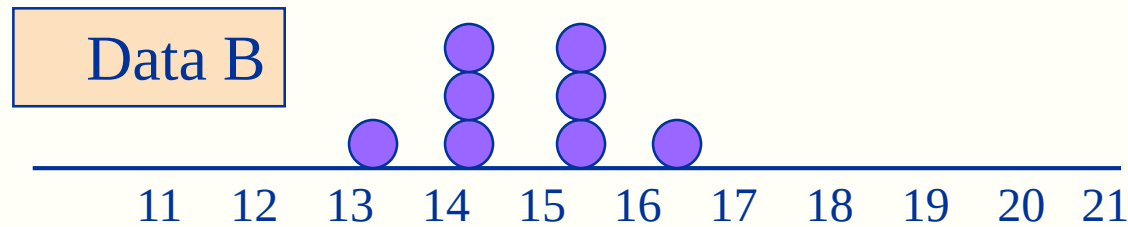
Measuring variation



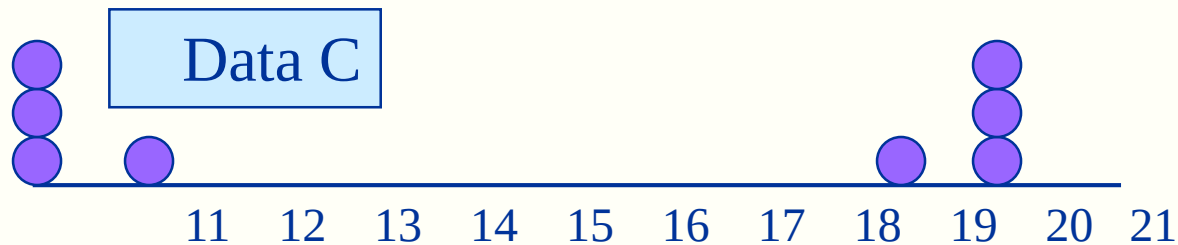
Comparing Standard Deviations



Mean = 15.5
 $s = 3.338$



Mean = 15.5
 $s = 0.926$



Mean = 15.5
 $s = 4.570$

Advantages of Variance and Standard Deviation

- ◆ Each value in the data set is used in the calculation
- ◆ Values far from the mean are given extra weight
(because deviations from the mean are squared)

Exercise

- ◆ A testing lab wishes to test two experimental brands of outdoor paint to see how long each will last before fading. The testing lab makes 6 gallons of each paint to test. Since different chemical agents are added to each group and only six cans are involved, these two groups constitute two small populations. The results (in months) are shown.
- ◆ **Brand A** 4 6 5 3
- ◆ **Brand B** 5 4 3 8
- ◆ Compute the following statistics
 - a) Range for brand A
 - b) Variance and standard deviation for brand B

Coefficient of Variation

- ◆ Measures **relative variation**
- ◆ Always in percentage (%)
- ◆ Shows **variation relative to mean**
- ◆ Can be used to compare two or more sets of data measured in **different units**

$$CV = \left(\frac{s}{\bar{x}} \right) \cdot 100\%$$

Comparing Coefficient of Variation

◆ City A (AA):

- Average Dialysis price = 4,800
- Standard deviation = 100

$$CV_{city A} = \left(\frac{S}{\bar{X}} \right) \cdot 100\% = \frac{100}{4800} \cdot 100\% = 2.08\%$$

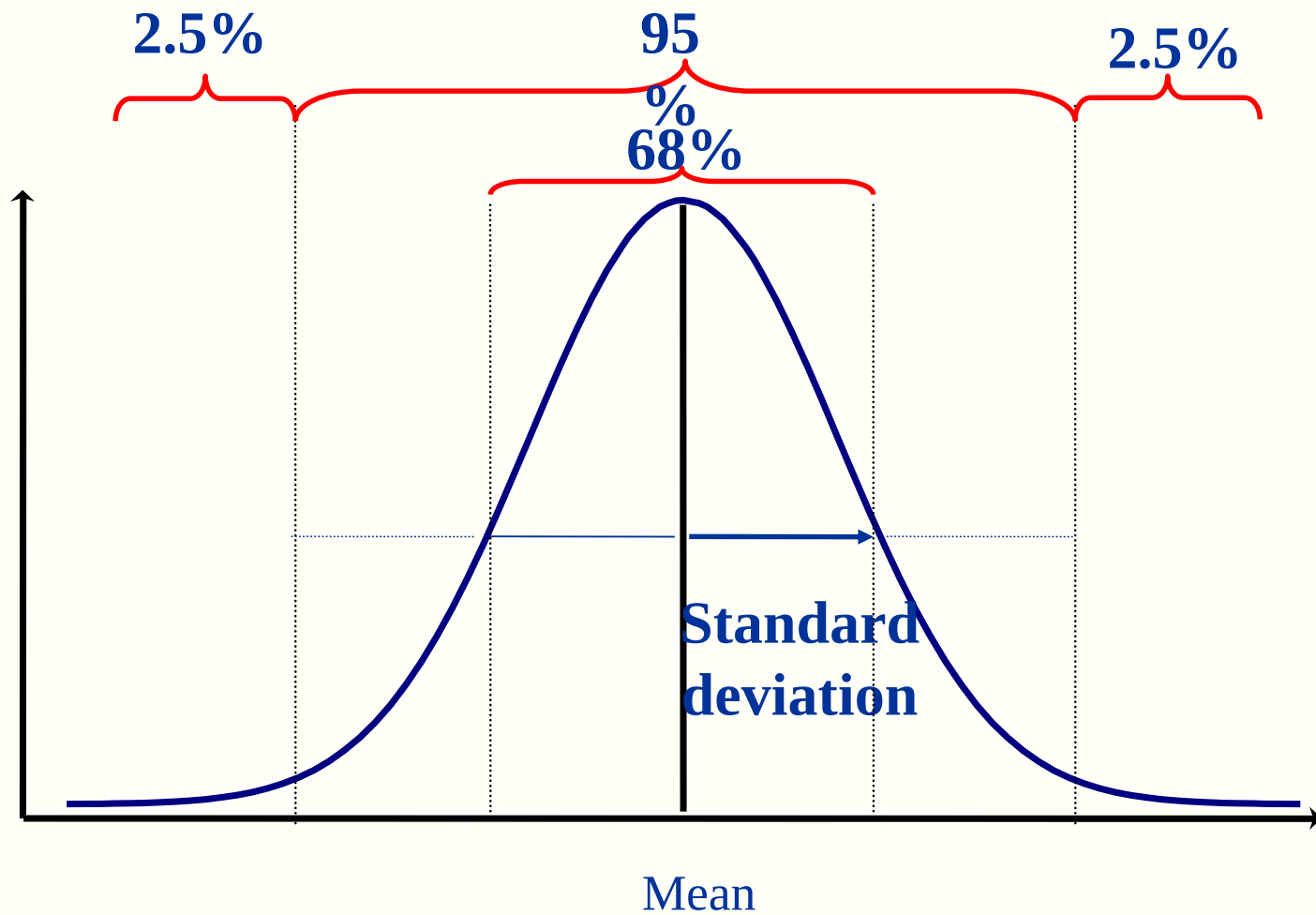
◆ City B (BahirDar):

- Average house price= 5,400
- Standard deviation = 100

Both cities have the same standard deviation, but city B is more variable relative to its price

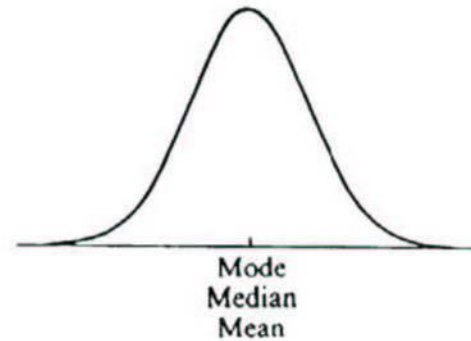
$$CV_{city B} = \left(\frac{S}{\bar{X}} \right) \cdot 100\% = \frac{100}{5400} \cdot 100\% = 1.85\%$$

Normal Distribution

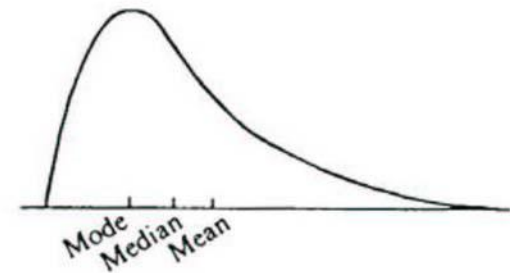


Comparison of Mode, Median and Mean

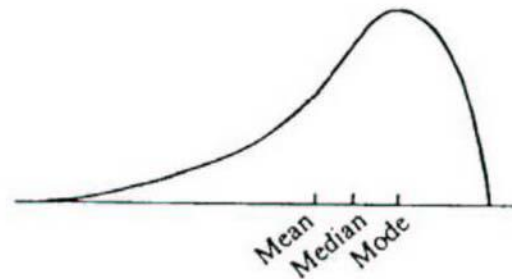
Symmetrical:
 $\text{Mode} = \text{Median} = \text{Mean}$



Skewed right:
 $\text{Mode} < \text{Median} < \text{Mean}$



Skewed left:
 $\text{Mean} < \text{Median} < \text{Mode}$



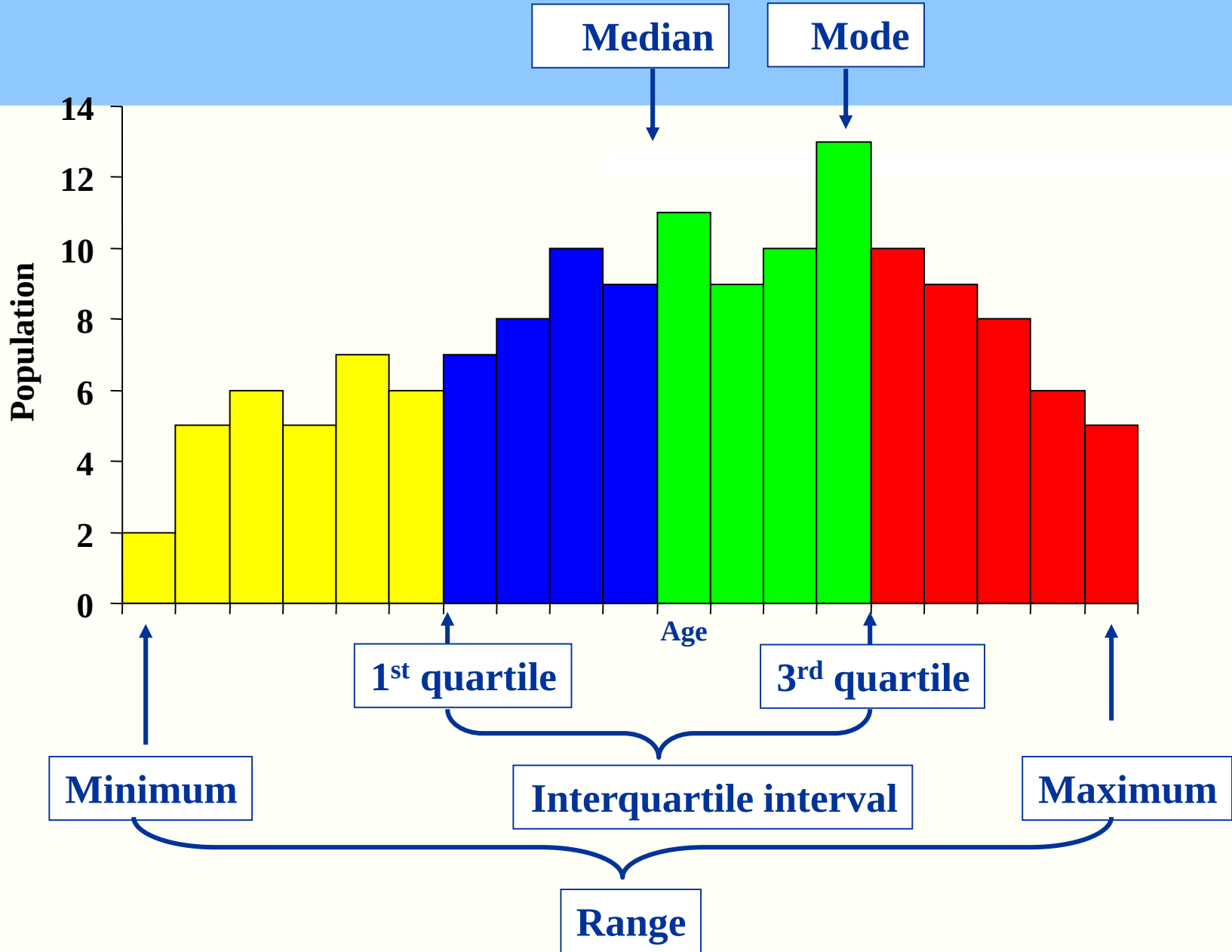
Name the Appropriate Measures of Central Location and Spread

Distribution	Central Location	Spread
Single peak, symmetrical	Mean*	Standard deviation
Skewed or Data with outliers	Median	Range or Interquartile range

* Median and mode will be similar

Properties of Measures of Central Location & Spread

- **Arithmetic mean – best for normally distributed data**
- **Median – best for skewed data**
- **Mode – simple, descriptive, not always useful**
- **Standard deviation – use with mean**
- **Range/Interquartile Range – use with median**



Seminar topics

- Skewness
 - Kurtosis
-

Thank you